The Evolution of Prices and Quantities of Occupational Human Capital *

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Abstract

I examine the roles of inter-cohort human capital growth and technical change in the rising U.S. wage inequality between 1970 and 2022, estimating the evolution of prices and quantities of human capital in occupational groups specializing in abstract, routine, and manual tasks. To quantify these changes, I use a flat spot price identification method that exploits wage changes for workers approaching retirement to infer shifts in prices of occupational human capital. Importantly, this method allows both the level and career accumulation profile of human capital to differ flexibly across cohorts. The results show that the price for abstract occupations has increased relative to manual and routine occupations, while prices for manual and routine occupations have declined in absolute terms. These price changes are consistent with the predicted role of technical change in increasing the relative demand for abstract tasks and decreasing the price of routine tasks. The relative wage growth of manual workers compared to routine workers has been driven primarily by the growth in the quantity of human capital linked to educational upskilling in manual occupations among younger cohorts rather than by an increase in the relative price of manual tasks.

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1 Introduction

Over the last five decades, changes in wage inequality in the U.S. have been characterized by diverging trends in the upper and lower tails of the wage distribution, with upper-tail inequality steadily rising and lower-tail inequality compressing (Acemoglu and Autor 2011). These varying patterns have become associated with changing wage differentials between workers employed in occupations with different task content (Autor, Levy, and Murnane 2003; Goos and Manning 2007; Acemoglu and Autor 2011). Specifically, wages have increased for workers in abstract task occupations at the top of the wage distribution and manual task occupations at the bottom, relative to those in routine task occupations in the middle.

A central challenge in interpreting trends in wage inequality is that observed hourly wages reflect two unobserved components: the market price of the occupational tasks workers perform, and the quantity of human capital workers supply to those tasks. Because neither is directly measured, changes in wage differentials can reflect shifts in the price of tasks, shifts in the human capital supplied to them, or both. Separately identifying these components is essential for understanding both the sources of wage inequality and the appropriate policy response. If stagnating wages reflect a decline in task prices driven by technological change, workers face labor market displacement that skill acquisition alone cannot offset, pointing toward policies that compensate for earnings losses directly, such as the wage insurance provisions of the Trade Adjustment Assistance program (Hyman, Kovak, and Leive 2024). If instead stagnating wages reflect a decline in the stock of human capital, due for example to deterioration in the quality of education or on-the-job training, supply-side interventions such as education and workforce training programs may be better suited to address the underlying skill deficit.

A leading explanation links wage polarization to changes in relative prices of occupational tasks driven by the decline in the cost of information technology, which substitutes for labor in routine tasks while complementing abstract and manual tasks (Autor, Levy, and Murnane 2003; Acemoglu and Autor 2011; Cortes 2016; Böhm 2020; Cavaglia and Etheridge 2020). Specifically, automation is predicted to lower the price of routine tasks while, through q-complementarity, raising the marginal productivity and prices of abstract and manual tasks

(Acemoglu and Autor 2011). Estimating the task prices needed to evaluate this mechanism is complicated, however, by shifts in the composition and quality of workers across cohorts and over the life cycle. As university enrollment expanded, college graduates increasingly filtered into lower-skill occupations compared to similarly credentialed workers in earlier decades (Beaudry, Green, and Sand 2016). Changes in workers' human capital across cohorts extend beyond compositional effects, encompassing shifts in university curricula and teaching methods and endogenous responses of human capital investment to technological shocks.[1]

In this paper, I reexamine the sources of rising wage inequality by decomposing wage changes into shifts in the relative prices of occupational human capital, which capture the potential effects of technical change operating through task prices, and shifts in the relative quantities of human capital, which reflect compositional and inter-cohort changes in workforce quality across occupations. Using data from the U.S. March Current Population Survey (MCPS) between 1970 and 2022, I estimate the evolution of the prices and quantities of human capital for abstract, routine, and manual occupational groups. I find that during this time period, the price for routine and manual occupations declined by 16.6 log points and 25.9 log points, respectively, while the price for abstract occupations increased by 8.9 log points. These price changes are consistent with the predicted role of technical change in decreasing the price of routine tasks and increasing the relative demand for abstract tasks.

By contrast, the rising wage premium of manual relative to routine workers is driven primarily by growth in the quantity of human capital in manual occupations. This suggests that supply-side changes offset any potential increase in the relative demand for manual tasks over this period. To understand the large inter-cohort increase in human capital

[1]For example, Branco, Carrillo, and Iglesias (2025) show that cohorts exposed to routine-biased technological change during their college-going years are more likely to complete a bachelor's degree, as automation reduces the opportunity cost of college attendance and raises the returns to higher education, generating systematic differences in human capital investment across cohorts. Carneiro and Lee (2011) and Hendricks and Schoellman (2014) find that an increase in college enrollment rates leads to significant changes in the average quality of college graduates' human capital. Bowlus and Robinson (2012) argue that the selection effect from the change in college enrollment rates was accompanied by technological improvement in the human capital production function for college graduate cohorts, improving the human capital quality of later cohorts of college graduates. Moreover, they show that accounting for the role of cohort effects significantly changes the estimates of human capital prices for college and high school graduates. Bastedo, Altbach, and Gumport (2016) highlight that university curricula significantly expanded in the late 20th century, driven by the growing influence of social movements and the differentiation of knowledge. Moreover, teaching and learning processes across established programs evolved in response to the information technology revolution.

among manual workers, I examine whether it reflects differences in human capital at labor market entry or differences in accumulation over the life cycle. Cohort-specific age profiles show little change in human capital at entry but steeper accumulation among workers born after 1950. I then investigate the role of educational upskilling. Holding the educational composition of manual occupations fixed at its 1970 level eliminates the increase in human capital across cohorts, indicating that the growth is primarily driven by the rising share of college-educated workers entering these occupations.

To estimate the price and quantity series, I adopt the flat spot identification method developed in Heckman, Lochner, and Taber (1998) and Bowlus and Robinson (2012). It is based on the prediction that the life-cycle human capital profile is concave, with a flat spot near the end of a worker's career during which changes in hourly wages are generated solely by price fluctuations (Ben-Porath 1967). A key advantage is that it allows for cohort effects in the production and distribution of human capital. I modify this method, originally developed for education-based human capital, by identifying different flat spot age ranges for workers in each occupational group and verifying that occupational switching within these ranges is limited. Because age, time, and cohort are collinear, identifying prices requires choosing an age range for the flat spot, and the estimates depend on that choice. I therefore report a benchmark specification alongside a range of alternative windows. The decline in routine and manual human capital prices and the growth in manual human capital quantity are robust in sign across these specifications, although their magnitudes vary.

By inferring shifts in the prices of occupational human capital from wage changes for workers in their flat spot age range, I can quantify how changes in both the prices and quantities of human capital contribute to the evolving wage inequality over the last five decades. Additionally, by focusing on workers' occupations rather than education, I provide evidence that technological change affects workers differently depending on the task content of their occupations, even if they have similar education levels, extending the findings of Bowlus and Robinson (2012) for educational human capital and providing an explanation for the decrease in the human capital price for college-educated workers. Therefore, my results emphasize that the task content of jobs provides relevant information beyond workers' education and supports the use of task-based models to analyze changes in wage inequality.

More generally, this paper contributes to the technical change literature by providing estimates of changes in the relative prices of occupational human capital while allowing both the initial level of human capital at labor market entry and the rate of accumulation over the career to be occupation-specific and to vary flexibly across cohorts.[2] While my estimates of changes in the relative prices of abstract occupations align with those of the previous literature, the estimates for manual occupations differ significantly, as wage growth driven by human capital accumulation across cohorts would otherwise be misattributed to price changes.[3] These findings highlight the importance of allowing the distribution of human capital to evolve flexibly across cohorts and over the life cycle when estimating long-run price changes.

I also examine how these trends extend to female workers, who experienced even larger shifts in occupational and educational composition over the sample period. Applying the male-estimated price series to female wages, I find that female workers experienced larger growth in human capital quantities than their male counterparts across all occupational groups. In routine occupations, male workers experienced a decline in hourly wages while female workers saw both wages and human capital levels increase. The contrast is most striking in manual occupations, where female workers experienced a dramatic increase in human capital quantities across cohorts that, despite the simultaneous decline in the price

[2]Previous approaches each restrict how human capital varies across cohorts, either in its distribution conditional on observed characteristics or in its life-cycle accumulation profile. Böhm (2020) uses repeated cross-sections with measures of pre-market talent and identifies price changes under the assumption that, conditional on those measures, the distribution of human capital is stable across cohorts. Cortes (2016) and Cavaglia and Etheridge (2020) use panel-data Roy-model approaches that absorb time-invariant worker-task heterogeneity through fixed effects and recover price changes from time-varying task effects, but do not allow cohort-specific life-cycle skill accumulation. Yamaguchi (2018) similarly decomposes wage changes into the prices and quantities of task-specific skills within a structural Roy model estimated on panel data, but identifies prices by assuming that the parameters of the skill-accumulation function are time-invariant. Böhm, von Gaudecker, and Schran (2024) allow occupation-specific, age-dependent skill accumulation and endogenous switching, yet still require expected skill growth to be time-invariant conditional on age and occupation histories.

[3]Specifically, when I focus on 1980–2010 rather than the full sample, for comparability with previous studies, I estimate a 24 log point increase in the price of human capital for abstract workers relative to routine workers, similar to Böhm (2020)'s estimate of a 25 log point increase between 1984–1992 and 2007–2009. By contrast, I find a small 1.8 log point increase in the price of human capital for manual workers relative to routine workers once I account for cohort effects, while Cortes (2016) estimates a relative price increase of 17% (15.7 log points) between 1976 and the mid-2000s. Taken together, the negligible relative price increase over 1980–2010 and the relative price decline over the full sample indicate that the relative wage growth of manual workers is not explained by growth in the relative price of manual human capital.

of manual human capital, translated into substantial wage growth. These patterns suggest that cohort effects in human capital accumulation are particularly pronounced for women, consistent with the rapid expansion of female labor market attachment and educational attainment over the last half-century (Goldin 2006; Bowlus and Robinson 2020).

The rest of the paper proceeds as follows. Section 2 explains how the flat spot price identification methodology is applied to occupational groups and motivates the use of the MCPS data. Section 3 presents the estimated price and quantity series, assesses their sensitivity to the flat spot assumption, examines the cohort dynamics behind the growth in manual human capital, and extends the analysis to female workers. Section 4 concludes.

2 Data and Methodology

2.1 Data and Summary Statistics

I obtain data on wages, labor supply, and occupational choices of U.S. workers from the MCPS (Flood et al. 2024), which provides the longest available data series on the social and economic indicators of the U.S. labor force. These employment data allow me to consistently control for the annual labor supply of workers and reduce issues concerning workers' selection into part-time jobs and retirement.[4]

Following Bowlus and Robinson (2012), I construct an hourly wage measure by dividing inflation-adjusted annual earnings by annual labor supply.[5] For the baseline analysis, I focus on the sample of male workers aged 25–64 years who are employed full time for the entire year, i.e., working 35-plus hours per week and 40-plus weeks per year.[6] Additionally, I exclude self-employed workers and workers without records of occupation, annual earnings, and variables

[4] I also use data from the Merged Outgoing Rotation Group (MORG) dataset, which provides measures of income and supply of labor. However, data on the number of weeks worked annually are available for only 12% of earners who report their earnings as an annual amount, which limits the potential to select individuals with strong labor market attachment.

[5] The annual labor supply is calculated as the product of weeks worked last year and hours typically worked per week last year since 1976 and as a product of weeks worked last year and hours worked last week before 1976. Prior to 1976, both weeks worked last year and usual hours per week last year are imputed from coarser MCPS measures; see Appendix A for details.

[6] In Subsection 3.4, I extend the analysis to female workers using an analogous sample construction.

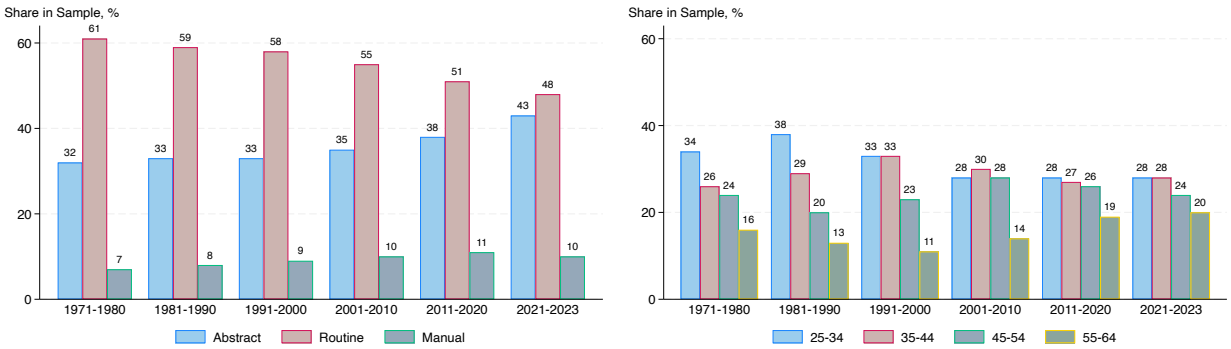
used to construct the measure of annual hours worked. To mitigate the influence of extreme values, I trim the top and bottom 1 percent of the hourly wage distribution in each year. Median hourly wages are used for the price series estimation to address the issue of top-coded values. While the MCPS data are available starting from 1964, the occupational classification scheme in the 1960s is not sufficiently detailed. Therefore, I focus on data beginning in 1971, when a more detailed occupational coding scheme is available.[7] Because the MCPS asks about earnings and weeks worked in the previous calendar year, I index wages and labor supply by the earnings year rather than the survey year. Thus, observations from the 1971–2023 March CPS correspond to earnings years 1970–2022.

I sort workers into abstract, routine, and manual occupational groups (Acemoglu and Autor 2011), based on the 1990 CPS occupational classification. This occupational grouping has been shown to preserve the relative ranking of occupational groups in terms of their task intensity and has been used consistently in the literature (Beaudry et al. 2016; Cortes 2016; Böhm 2020; Cavaglia and Etheridge 2020).[8] The abstract occupational group includes workers in managerial, professional, and technical occupations. The routine occupational group includes workers in sales, clerical, and administrative occupations and production, crafts, repair, and operative occupations. Finally, the manual occupational group includes workers in service occupations.

Figure 1 depicts the age and occupational composition of the sample. Panel (a) illustrates the polarization of the labor market's occupational structure, with an increasing share of abstract and manual occupations and a declining share of routine occupations. Despite this trend, workers in routine occupations remain the largest group in the sample, while those in manual occupations form the smallest. Panel (b) highlights an aging trend in the sample over time, consistent with U.S. demographic patterns. Over the decades, the share of workers aged 25–34 declines, that of workers aged 35–44 fluctuates between 26% and 33%, while

[7] To make occupational groups comparable across time, I use a cross-walk between the 1990 occupational classification and the classification used in each survey year, constructed by the Integrated Public Use Microdata Series (https://doi.org/10.18128/D030.V12.0). The results are robust to using an alternative cross-walk based on Autor and Dorn (2013).

[8] In Appendix B, I analyze the average task intensity across occupational groups based on Dictionary of Occupational Titles data. I find that workers in each occupational group tend to specialize in tasks according to their classification. For example, the abstract occupational group has the highest average intensity of abstract tasks.



(a) Occupation (b) Age

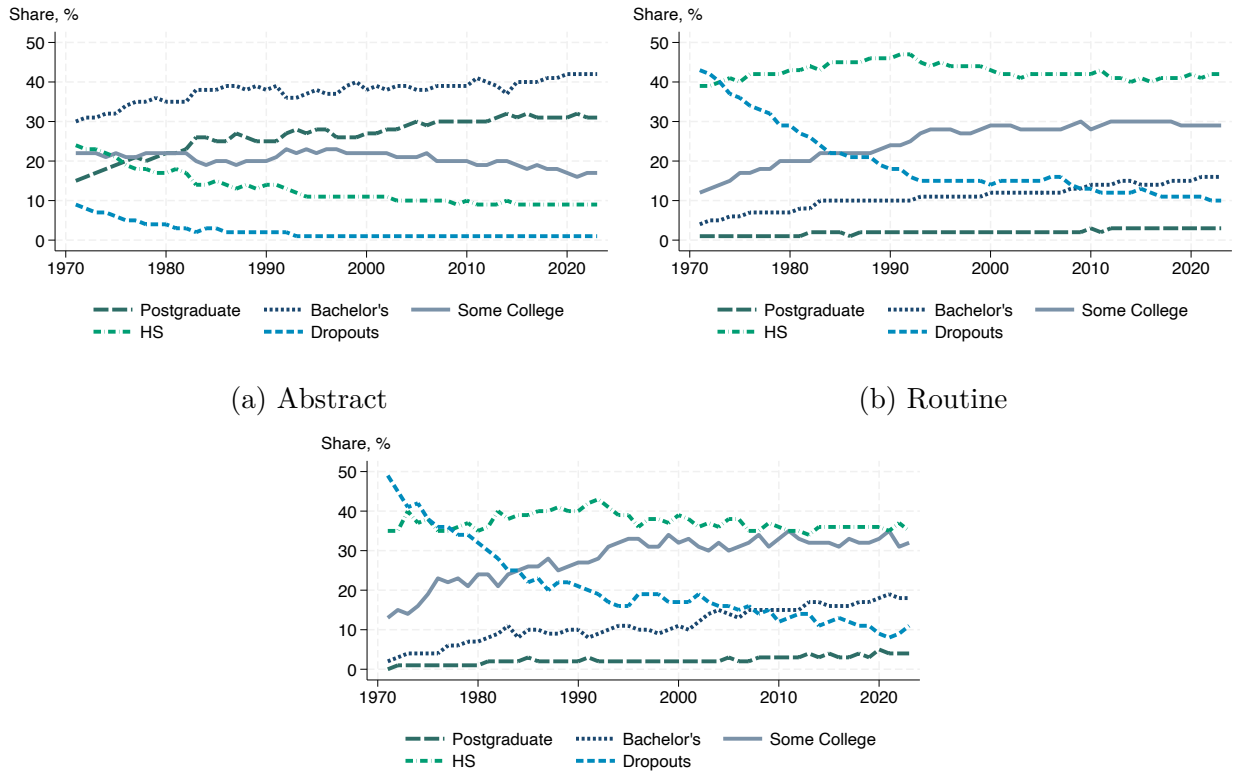
Figure 1: Sample Composition by Decade

Notes: This figure shows summary characteristics for the sample of full-time, full-year male wage and salary workers aged 25–64 years in the MCPS, 1971–2023. The sample excludes self-employed workers and workers without records of occupation, annual earnings, and annual hours worked. Panel (a) summarizes the sample's occupational composition, while panel (b) summarizes its age composition. All means and shares are computed using CPS sampling weights.

those aged 45–54 and 55–64 together account for an increasing share of the sample, rising from 33% in the 1980s to 44% in the 2020s.

Figure 2 illustrates the evolution of the educational composition of workers in abstract, routine, and manual occupations. Abstract occupations have been increasingly performed by the most educated group of workers, while less-educated workers have been crowded out of abstract occupations into manual and routine occupations. As college attainment has expanded, workers with some college education or bachelor's degrees have increasingly sorted into routine and manual occupations, reflected in a lower share of high school dropouts and graduates, whose presence in these occupations has declined. These compositional shifts, together with the transformation of university curriculum and learning, imply that the distribution of human capital has undergone substantial changes since the 1970s (Bowlus and Robinson 2012; Bastedo, Altbach, and Gumport 2016). Because these changes reflect differences in the educational endowments carried into the labor market by successive birth cohorts, which are examined more closely in Subsection 3.3, cohort effects are central to the price-estimation framework.

Figure 3 shows the evolution of the log median hourly wages for workers in abstract, routine, and manual occupations. The trends in the log median wage reflect the expansion of



(a) Abstract

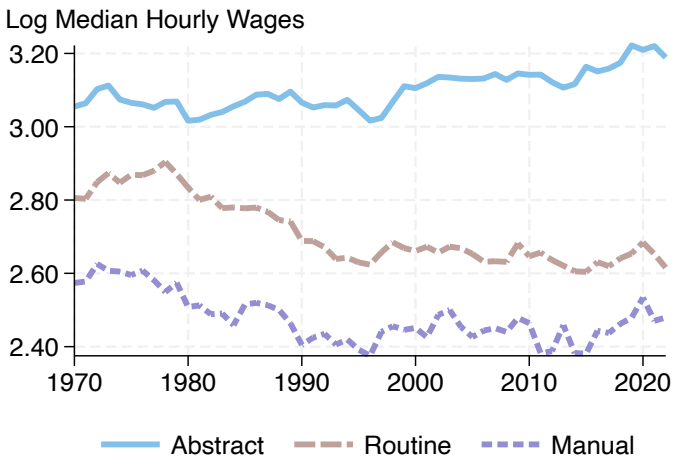
(b) Routine

(c) Manual

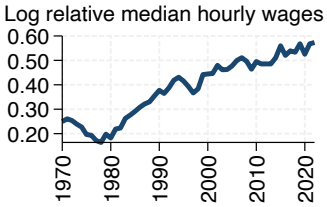
Figure 2: Educational Composition by Occupational Group over Time

Notes: This figure shows the evolution over the sample period of the educational composition of workers in abstract, routine, and manual occupational groups in the sample of full-time, full-year male wage and salary workers aged 25–64 years in the MCPS. The sample excludes self-employed workers and workers without records of occupation, annual earnings, and annual hours worked. All means and shares are computed using CPS sampling weights.

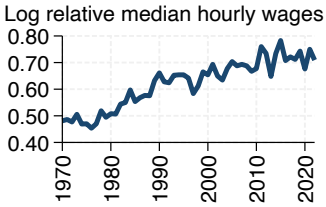
wage inequality in the upper part of the wage distribution and the compression of inequality in the lower part of the wage distribution (Acemoglu and Autor 2011; Böhm 2020). The log median hourly wage increases for abstract workers and decreases for manual and routine workers. These patterns generate an increase in the wage premium for abstract occupations relative to routine and manual occupations. Routine workers show a larger decline in the log median wage than manual workers, and therefore the wage premium for routine occupations relative to manual occupations declines.



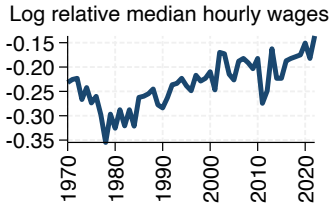
(a) All Occupational Groups



(b) Abstract vs. Routine



(c) Abstract vs. Manual



(d) Manual vs. Routine

Figure 3: The Evolution of the Log Median Hourly Wage

Notes: Panel (a) shows the evolution of the log median hourly wages in abstract, routine, and manual occupations for the sample of full-time, full-year male wage and salary workers aged 25–64 years in the MCPS. Panels (b)–(d) show the evolution of relative wages. All median hourly wages are computed using CPS sampling weights.

2.2 Methodology

I model a worker's hourly wage as the product of the efficiency units of human capital the worker supplies to the occupational group and the competitive equilibrium price per efficiency unit paid in that group. I refer to this price as the *price of occupational human capital*. I sometimes use *task price* as shorthand, where that interpretation is exact only when each broad occupational group is treated as a distinct composite task in production. This aggregation is common in grouped-occupation implementations of RBTC.[9] Formally,

[9] Acemoglu and Autor (2011) use broad occupational categories as task proxies, Cortes (2016) models three occupations and three tasks, Cavaglia and Etheridge (2020) estimates prices for occupations grouped by predominant tasks, and Böhm (2020) estimates prices for broad occupational groups treated as tasks.

the period t hourly wage for worker i of age a employed in occupational group $o \in \{A, R, M\}$, where A, R, and M denote groups specializing in abstract, routine, and manual tasks respectively, is the product of the task price P_t^o and the efficiency units of human capital $H_{i,t}^{o,a}$ that the worker supplies to the task:

$$W_{i,t}^{o,a} = P_t^o \times H_{i,t}^{o,a}. \tag{1}$$

Equation (1) embeds the key identifying assumption: a single price P_t^o applies to every efficiency unit supplied to group o, so all workers in the group are paid the same price per efficiency unit regardless of their education, cohort, or age.[10] Therefore, the observed changes in wages and wage inequality can be driven either by shifts in task prices or by changes in the human capital workers supply to those tasks, neither of which are directly observed in the data. Moreover, different cohorts of workers likely differ in their human capital distribution due to dramatic shifts in educational composition across time and technological and methodological advancements in education.[11] Nevertheless, it is important to separately identify the evolution of task prices and human capital quantities over time because they have different implications for the underlying causes of changing wage inequality.

I identify task price sequences by applying the flat spot price identification method of Heckman, Lochner, and Taber (1998) and Bowlus and Robinson (2012). This method builds on the Ben-Porath (1967) human capital investment model, which predicts that gains from human capital investment decline as workers approach retirement. Consequently, the life-cycle human capital profile exhibits a concave shape with a flat spot prior to retirement, around which human capital levels are constant. Under this identification assumption, work-

[10] Within a group, workers and cohorts differ in the quantity $H_{i,t}^{o,a}$ they supply, not in the price they receive. Accordingly, P_t^o is the human capital price of Bowlus and Robinson (2012) applied to occupational groups rather than schooling groups, and shifts in a group's educational composition, including educational upskilling, operate through $H_{i,t}^{o,a}$ rather than P_t^o.

[11] Carneiro and Lee (2011) and Hendricks and Schoellman (2014) show that higher college enrollment rates significantly alter the average quality of college graduates' human capital, while Bowlus and Robinson (2012) and Bastedo, Altbach, and Gumport (2016) highlight that the production of human capital in universities has changed over time. Attanasio, Blundell, Conti, and Mason (2020) show that even the distribution of human capital in early childhood has changed for cohorts born decades apart.

ers in their flat spot age range, a^*, have stable levels of human capital,

$$E[\ln H_{i,t}^{o,a^*} - \ln H_{i,t-1}^{o,a^*-1}] = 0, \tag{2}$$

and the wage growth during the flat spot age range reflects changes in the task price,

$$E[\ln W_{i,t}^{o,a^*} - \ln W_{i,t-1}^{o,a^*-1}] = \ln P_t^o - \ln P_{t-1}^o. \tag{3}$$

I use this identification assumption to identify the task price change from the average wage growth of workers in the flat spot age range of their human capital profile.

To implement the flat spot method using the MCPS data, I estimate median hourly wages, task prices, and efficiency unit series in each occupational group. Following Bowlus and Robinson (2012), I use median rather than mean wages to address the time-varying treatment of top-coded earnings values in the MCPS.[12] Moving from the mean-based condition in (2) to a median-based estimator requires the additional assumption that the distribution of human capital supplied to tasks is stable within the synthetic cohort across the flat spot age range, that is, that workers aged a^* in year t and workers aged $a^* - 1$ in year $t - 1$, who belong to the same birth cohort, have the same distribution of human capital. This assumption follows naturally from the flat spot condition itself: if human capital investment has ceased for this cohort, the entire distribution of human capital levels, and hence the median, remains unchanged. Under this additional assumption, the log price change can be identified from median wages:

$$\text{Med}[\ln W_{i,t}^{o,a^*}] - \text{Med}[\ln W_{i,t-1}^{o,a^*-1}] = \ln P_t^o - \ln P_{t-1}^o. \tag{4}$$

I estimate the price and efficiency unit series in three steps. First, the sample median hourly wage is computed for each age-year cell within each occupational group. For each flat spot age a, I compute the increase in the log median wage between workers of age a in period t and workers of age $a - 1$ in period $t - 1$. The log price change between t and $t - 1$ is the average of these increases across all flat spot ages. The cumulative price level in year t is obtained by summing all annual log price changes from the base year. Second,

[12] See Appendix C for a derivation of this decomposition. Appendix D shows that the estimated price series are robust to computing wages as means or as means of logs rather than medians.

the log median hourly wage in year t for each occupational group is computed as the log of the median hourly wage across all workers aged 25–64 in that group. Finally, the log efficiency units of human capital in year t are computed as the difference between the log median hourly wage and the log price of human capital.[13]

Applying the flat spot price estimation method to occupational human capital requires identifying the baseline flat spot age range for workers in each occupational group. To identify the plausible flat spot region, I draw on two complementary sources of evidence. First, I build on the relationship between labor market entry age and flat spot age range for workers with varying levels of education documented in Bowlus and Robinson (2012) and exploit the educational composition of workers and compositional shifts across cohorts. Second, I use the direction of inter-cohort changes in human capital implied by observable patterns in occupational selection and cross-sectional wage profiles to place bounds on the flat spot age range.

The logic behind using cross-sectional profiles to place bounds on the flat spot age range follows from the relationship between cohort effects and the slope of observed age-earnings profiles. In any given year, the slope of the cross-sectional age-earnings profile within an occupational group reflects two opposing forces: an experience effect, capturing how human capital grows as workers age, and a cohort effect, capturing how the initial human capital of workers observed in the occupation differs across successive cohorts (Fang and Qiu 2023). In an occupational setting, this cohort effect absorbs both differences in the human capital of labor market entrants and differences in occupational selection across cohorts. When cohort effects are positive, that is, when younger cohorts enter with higher human capital than older ones, the cross-sectional profile peaks at younger ages than the true life-cycle human capital profile, since the experience effect must overcome the downward pull of cohort differences for the profile to slope upward. Because the flat spot begins no earlier than the peak of the life-cycle human capital profile, and the cross-sectional hourly wage profile peaks earlier than the life-cycle profile under positive cohort effects, the cross-sectional peak must occur

[13]The confidence intervals for the estimated annual price changes are reported in Appendix E. To construct these, I estimate median hourly wages via quantile regression for each age-year cell within each occupational group, pooling two adjacent years. The delta method is then applied to the resulting coefficients to account for the nonlinear transformation into log median wages and price changes.

before the flat spot begins. Therefore, the cross-sectional peak provides a lower bound on the flat spot age range. I apply this reasoning separately to each occupational group below, with the full identification argument developed in Appendix F.

For manual occupations, both the educational composition of workers and cohort-level patterns in occupational selection support a flat spot age range of 46–55. The average level of schooling for workers in manual occupations over the sample period slightly exceeds 12 years (see Appendix Figure F.1), and the occupation has a high share of high school graduates (see Figure 2), placing the flat spot age range in line with that of high school graduates in Bowlus and Robinson (2012). The share of workers employed in manual occupations has been steadily increasing across cohorts (see Appendix Figure F.2). Under a one-dimensional model of ability sorting along the occupational hierarchy, in which manual occupations are the lowest-skill group (Acemoglu and Autor 2011; Autor and Dorn 2013), the expansion of employment in manual occupations implies that marginal workers entering these occupations have higher ability than the average incumbent, generating positive ability selection effects.[14] These patterns are reinforced by improvements in the human capital production function for college-educated workers documented in Bowlus and Robinson (2012), whose presence in manual occupations has grown across cohorts. Together, they imply a positive aggregate cohort effect for manual workers. Under positive cohort effects, the flat spot must occur later than the peak of the cross-sectional wage profile. Cross-sectional wage profiles for manual workers continue to grow throughout the early 40s for all decades beginning in the 1980s (see Appendix Figure F.4), confirming that the human capital profile cannot peak before the mid-40s and supporting the flat spot range of 46–55.[15]

For routine occupations, the direction of cohort effects is theoretically ambiguous, as these occupations face simultaneous outflows to both higher-skill abstract and lower-skill

[14]By contrast, Böhm et al. (2024) document that a multidimensional skill model better fits German longitudinal data, where entrants into all occupations, including low-wage ones, earn less than incumbents. However, since I do not observe occupational tenure and estimate a single aggregate human capital level per occupational group, my approach does not distinguish between skill dimensions within an occupation. The one-dimensional sorting assumption is therefore consistent with the level of aggregation at which human capital is identified in this paper. The U.S. evidence on ability patterns of occupational switching in Cortes (2016) further supports this assumption for the same occupational classification and an overlapping time period.

[15]The length of the flat spot region is set to 10 years to generate a reasonable sample size, similar to that used in Bowlus and Robinson (2012). Sensitivity to the choice of window length is reported in Table 1.

manual jobs (Cortes 2016), making it difficult to sign the net ability selection effect. I therefore draw on two complementary arguments to support the same flat spot range of 46–55. First, the educational attainment of routine workers closely mirrors that of manual workers, with average schooling slightly exceeding 12 years throughout the sample period (see Appendix Figure F.1), suggesting a similar timeline of labor market entry for workers in this group. Second, the nature of routine tasks provides a lower bound on the flat spot age range: unlike manual occupations, which rely heavily on physical capabilities that deteriorate earliest, routine tasks are predominantly cognitive or procedural in nature, implying that human capital depreciation in routine occupations is unlikely to begin earlier than in manual occupations. Cross-sectional wage profiles for routine workers also remain upward-sloping through the early 40s for all decades beginning in the 1980s (see Appendix Figure F.4), further confirming that the human capital profile cannot peak before the mid-40s. Together, these arguments support the flat spot range of 46–55 for routine workers.

In contrast to routine and manual occupations, abstract occupations are dominated by college-educated workers throughout the sample period, with workers holding at least a bachelor's degree consistently constituting the largest educational group (see Figure 2). This higher average educational attainment implies that human capital accumulation continues later into the life cycle for abstract workers, placing the plausible flat spot range around ages 50–59, the flat spot for college graduates in Bowlus and Robinson (2012). To verify this range, I examine cohort-level patterns in occupational selection. For cohorts born between 1950 and 1956, the share of workers in abstract occupations in their late 30s, as well as the share of college graduates, decreased (Appendix Figure F.2), implying positive cohort effects for these cohorts. Under positive cohort effects, the flat spot must occur later than the peak of the cross-sectional wage profile. Cross-sectional wage data from 2001, when the 1950 cohort was 51 and the 1956 cohort was 45, reveal that wages for abstract workers peaked in the late 50s (see Appendix Figure F.3). This leads me to select 51–60 as the flat spot age range for the baseline results, one year later than the flat spot identified for college graduates in Bowlus and Robinson (2012), reflecting the slightly older peak earnings age in abstract occupations.[16]

[16]While for manual workers I set the flat spot to begin just after the cross-sectional profile stops rising,

These benchmark ranges, 46–55 for manual and routine workers and 51–60 for abstract workers, rest on an assumption about where the life-cycle human capital profile is flat. Because the collinearity of age, time, and cohort makes some such assumption necessary, I explore the sensitivity of the estimates to it across a range of alternative flat spot specifications. Table 1 reports the benchmark estimates together with the range obtained across these specifications.

2.3 Discussion of the Methodology

Having established the flat spot age ranges for each occupational group, I now turn to discussing the nature of the occupational human capital identified by the flat spot approach. I define the type of occupational human capital by the task content of workers' occupations rather than by their educational attainment. Workers employed in abstract, routine, and manual occupational groups are assumed to supply the same *type* of human capital within each group, that is, they specialize in the same broad category of tasks, while differing in the *quantity* of human capital they supply to those tasks depending on their education, ability, and age. This approach extends the canonical model of wage inequality (Katz and Murphy 1992; Goldin and Katz 2008), which classifies workers by educational attainment (typically college versus non-college) and treats the human capital of each group as a single homogeneous input. In the canonical framework, a college graduate always supplies high-skill services regardless of occupation. Instead, this approach connects to the task-based framework commonly applied in the routine-biased technical change (RBTC) literature (Autor, Katz, and Kearney 2006; Acemoglu and Autor 2011; Cortes 2016), in which skills and tasks are conceptually distinct: workers with different levels of education and ability can perform the same type of occupational task while supplying different quantities of human capital to it.

My measure of occupational human capital is the quantity of efficiency units that a worker supplies to the broad task content of the occupational group. Education, ability, ex-

applying the same rule for abstract workers would place the flat spot above age 57, where non-random selection into retirement among older high-skilled workers is more likely to contaminate wage growth estimates. Setting the flat spot to begin at 51 therefore balances the lower bound implied by the cohort effect argument against the upper bound imposed by retirement selection.

perience, occupation-specific tenure, and other worker characteristics may all contribute to this quantity, but the method does not separately identify their contributions. Accordingly, the estimated price is the price of an efficiency unit of this broad occupational input. Comparing this price over time requires the composition of an efficiency unit to remain stable across cohorts and periods. The flat-spot restriction then requires the median quantity of these efficiency units supplied by a synthetic cohort to remain approximately constant over the selected age range.

The flat spot approach has an important advantage over these alternatives for estimating price changes over long horizons because it allows both the initial level of human capital at labor market entry and the rate of accumulation over the career to be occupation-group-specific and to vary flexibly across cohorts. The Roy model and propensity-based methods discussed in the introduction instead impose some form of cohort invariance, either in the human capital accumulation profile (Cortes 2016; Cavaglia and Etheridge 2020; Yamaguchi 2018; Böhm et al. 2024) or in a stable distribution of pre-market talents (Böhm 2020). While these assumptions might hold in the short run, they are unlikely to hold over the half-century covered by my analysis, given that human capital investment has responded to technological change differently across successive cohorts (Branco et al. 2025). Moreover, the presence of cohort effects in the human capital production function documented by Bowlus and Robinson (2012) and Carneiro and Lee (2011) would confound long-run price change estimates if not accounted for.

The flat-spot method identifies changes in the price of occupational human capital, up to a normalization, under three conditions. First, within each birth-cohort and occupational-group cell, supplied human capital must be approximately stable over the flat-spot age range. In my empirical implementation, which is based on median wages, this condition is best understood as stability in the median supplied human capital over those ages. Second, workers within each broad occupational group must be interpretable as supplying the same broad type of human capital, or at least a stable broad task bundle, even if they differ in the quantity supplied. While this is an identifying approximation, Appendix B shows that workers in each group remain concentrated, on average, in the expected broad task category over the flat-spot ages. Third, wage changes over the flat-spot range must mainly reflect

shifts in the market price of that input rather than deferred compensation or other employer-specific contractual wage dynamics. Evidence on age profiles of non-wage compensation across occupations suggests that such contractual differences are unlikely to be the main source of the estimated price changes (see Appendix G).

One additional issue associated with applying the flat spot price identification method to occupation-based human capital relates to occupational switching and the stability of skill groups defined by occupations. The wage growth of workers in their flat spot age range in a synthetic cohort reflects the change in the price of occupational human capital only if composition effects due to workers switching occupations are small. Using the panel component of MORG files, I find that the large majority of workers in their flat spot age range remain in the same occupational group over a year, with one-year persistence of roughly 85% or more for abstract and routine workers and close to 80% for manual workers. This share remains high across the flat spot age range and over time (see Appendix H). Occupational choice is endogenous: workers sort into occupations based on the task prices they face and the quantity of human capital they can supply. The broad definition of occupational groups used here, with each group encompassing multiple occupations intensive in abstract, routine, or manual tasks, limits the role of switching for price identification, since most occupational mobility occurs within rather than between these broad groups. Consistent with this, comparing the price series for a sample of full-time MORG workers with the sample restricted to occupation stayers shows that the effect of occupational switching on the estimated price series is limited (see Appendix I). Other studies have also shown that the role of selection in occupation stayers decreases with age as match quality increases with experience.[17]

[17] Gathmann and Schönberg (2010) find that the frequency and distance of occupational changes decline with age. Cavounidis and Lang (2020) show that in a dynamic skill formation model, workers' responses to wage shocks decline with age because they face a shortened horizon of future earnings and are more heavily invested in their existing stock of skills.

3 Results

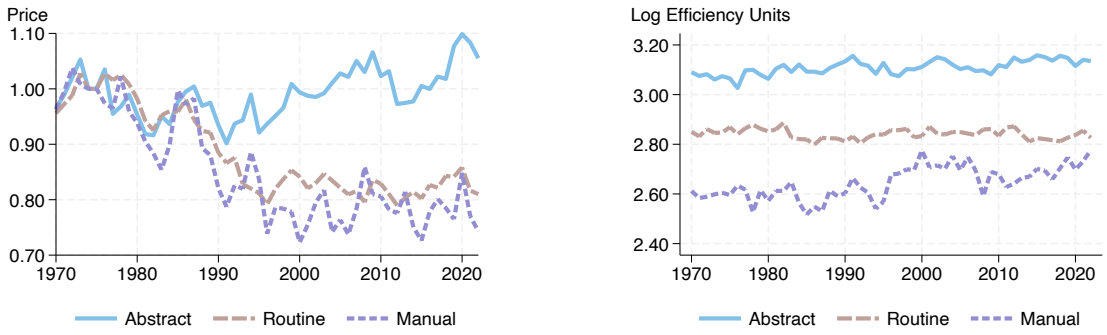
3.1 Price and Quantity Series

Figure 4 presents the estimation results for the prices and quantities of human capital in abstract, routine, and manual occupations over 1970–2022. Panel (a) shows the estimated price series.[18] The prices for all occupations tend to fluctuate and drop during the recessions of the mid-1970s, early 1980s, early 1990s, and 2020–2022. However, clear trends emerge: the price series for manual and routine workers decline over the observed period and are highly correlated ($\rho = 0.9$), while the price series for abstract workers remain relatively stable. Panel (b) shows the evolution of log efficiency units of occupational human capital levels per worker. While the quantity of human capital in abstract and routine occupations has changed little, the quantity in manual occupations has grown substantially.

Table 1 reports the corresponding changes between 1970 and 2022. It is organized in three panels, for log wage, log price, and log human capital changes. Within each panel, columns (1)–(3) report changes for abstract, routine, and manual occupations, while columns (4) and (5) report changes relative to routine occupations; each row corresponds to a flat spot specification, with the baseline in the first row. Wage growth is actual median wage growth and does not vary across specifications, so the wage panel has a single row. Under the baseline specification, the price of abstract human capital increased by 8.9 log points, while the prices of routine and manual human capital declined by 16.6 and 25.9 log points. Human capital per worker moved in the opposite direction for manual workers, growing by 16.5 log points, against 4.5 log points in abstract occupations and a decline of 2.4 log points in routine occupations. Because wage growth is the sum of the two components, the 32.4 log point gain in abstract wages relative to routine wages is driven mainly by the relative price, which rose by 25.5 log points, while the 9.6 log point gain in manual wages relative to routine wages comes entirely from relative human capital, which rose by 18.9 log points against a 9.3 log point fall in the relative price.

Because age, time, and cohort are collinear by construction, estimating the relative price

[18] The price must be normalized for one of the years. I also normalize prices in 1974 and 1975 to 1 due to the change in the MCPS' reporting of weekly hours worked.



(a) Price Series

(b) Log Efficiency Unit Series

Figure 4: The Evolution of the Prices and Quantities of Occupational Human Capital

Notes: This figure shows the evolution of estimated prices and quantities of human capital in abstract, routine, and manual occupational groups in the sample of full-time, full-year male wage and salary workers in the MCPS. The estimation uses median hourly wages computed with CPS sampling weights.

of human capital requires an identifying assumption about the shape of the age-earnings profile, and the flat spot age range is therefore a choice rather than an estimate. I consequently report results under a range of alternative flat spot specifications. Table 1 presents estimated price and quantity changes when the flat spot is shifted 2 or 4 years earlier or 2 years later than the benchmark, shifted earlier only for routine and manual occupations, shortened from 10 to 5 years while holding the start ages fixed, and shortened only for routine and manual occupations. The shifts address uncertainty about when the flat spot begins while the shorter windows address how long it persists. According to the theory of human capital, the choice of flat spot age range affects price estimates in predictable ways. Choosing a flat spot that is too early will result in upward-biased price change estimates, since the estimated price changes will absorb wage growth due to human capital accumulation; setting it too late will result in downward-biased estimates, since it will absorb the effect of human capital depreciation.[19] The estimates in Table 1 are broadly consistent with this prediction: shifting the flat spot earlier generally yields higher (less negative) price change estimates, and shifting it later generally yields lower (more negative) estimates.

Columns (4) and (5) of Table 1 report the relative changes across all specifications.

[19]A similar prediction applies to the length of the window. If human capital begins to depreciate before the assumed flat spot ends, shortening the window by removing the oldest ages should produce a higher estimated price change.

Relative prices drive the abstract-to-routine comparison, while relative human capital drives the manual-to-routine comparison. The relative price of abstract to routine human capital rises in every specification, by between 11.8 and 36.8 log points against a relative wage increase of 32.4 log points, and accounts for more of that gain than the quantity change does in all but one specification. The relative quantity of manual to routine human capital rises in every specification, by between 8.2 and 27.1 log points against a relative wage increase of 9.6 log points. Although the individual estimates vary in magnitude, these results broadly support the baseline interpretation.

For comparison with the time periods studied in the previous literature, I focus on changes in relative prices over 1980–2010 rather than over the full 1970–2022 sample period.[20] Under the baseline specification, the price for abstract workers relative to routine workers increases by 24 log points between 1980 and 2010, similar to both the 25 log point increase in the price of abstract human capital relative to routine human capital between 1984–1992 and 2007–2009 estimated in Böhm (2020) and the 25 log point increase in the price of abstract human capital relative to manual human capital between 1976 and the mid-2000s estimated in Cortes (2016). By contrast, I find a negligible 1.8 log point increase in the price for manual occupations relative to routine occupations over the same period. This finding contrasts with the significant increase in the price for manual occupations relative to routine occupations documented by Böhm (2020) and Cortes (2016), who estimate that the relative price increased by 32.9 log points between 1984–1992 and 2007–2009 and by 17% (15.7 log points) between 1976 and the mid-2000s, respectively.

While the exact source of this discrepancy is difficult to isolate, as these studies differ from mine in their datasets and identification strategies, one important explanation lies in the treatment of cohort effects. The flat spot price estimation method allows for cohort effects in the distribution of human capital supplied by workers as well as the shape of the occupation-group-specific human capital profile, restricting only that human capital is stable within the flat spot age range near retirement. Without controlling for cohort effects, changes in wage differentials between occupations that are driven by shifts in the level or

[20]These changes are not reported in Table 1 but correspond to changes in the relative price series plotted in Figure 5.

Table 1: Robustness of Estimated Changes in Prices and Human Capital Quantities to Flat Spot Assumptions

	Flat spot			(1)	(2)	(3)	(4)	(5)
	Abstract	Routine	Manual	Abs	Rt	Man	Abs−Rt	Man−Rt
Log wage change, 1970–2022 (log points)								
(all specifications)				13.4	-19.0	-9.4	32.4	9.6
Log price change, 1970–2022 (log points)								
Baseline	51–60	46–55	46–55	8.9	-16.6	-25.9	25.5	-9.3
4 years earlier	47–56	42–51	42–51	33.3	-2.9	-15.2	36.2	-12.3
2 years earlier	49–58	44–53	44–53	29.4	-7.4	-13.6	36.8	-6.2
2 years later	53–62	48–57	48–57	0.0	-27.6	-45.1	27.6	-17.5
2 years earlier for Rt/Man	51–60	44–53	44–53	8.9	-7.4	-13.6	16.3	-6.2
4 years earlier for Rt/Man	51–60	42–51	42–51	8.9	-2.9	-15.2	11.8	-12.3
4 years earlier for Man only	51–60	46–55	42–51	8.9	-16.6	-15.2	25.5	1.4
5-year window	51–55	46–50	46–50	24.3	-9.8	-23.0	34.1	-13.3
5-year window for Rt/Man	51–60	46–50	46–50	8.9	-9.8	-23.0	18.7	-13.3
Log human capital change, 1970–2022 (log points)								
Baseline	51–60	46–55	46–55	4.5	-2.4	16.5	6.9	18.9
4 years earlier	47–56	42–51	42–51	-19.9	-16.1	5.8	-3.9	21.8
2 years earlier	49–58	44–53	44–53	-16.0	-11.6	4.2	-4.5	15.8
2 years later	53–62	48–57	48–57	13.4	8.6	35.7	4.8	27.1
2 years earlier for Rt/Man	51–60	44–53	44–53	4.5	-11.6	4.2	16.1	15.8
4 years earlier for Rt/Man	51–60	42–51	42–51	4.5	-16.1	5.8	20.6	21.8
4 years earlier for Man only	51–60	46–55	42–51	4.5	-2.4	5.8	6.9	8.2
5-year window	51–55	46–50	46–50	-10.9	-9.2	13.6	-1.7	22.8
5-year window for Rt/Man	51–60	46–50	46–50	4.5	-9.2	13.6	13.7	22.8

Notes: All entries are log point changes between 1970 and 2022. Wage growth is actual median hourly wage growth and does not vary across flat spot specifications. Price changes are estimated using the flat spot method with the indicated adjustment to the baseline flat spot ages (Abstract: 51, Routine: 46, Manual: 46) and window length. Human capital change is the residual quantity change (wage growth minus price change). The baseline flat spot window length is 10 years; the shorter-window specifications hold the start ages fixed and truncate the window to 5 years from the top. The estimation uses median hourly wages computed with CPS sampling weights.

accumulation profile of human capital across cohorts would instead be attributed to changes in the price of human capital. This is particularly evident for workers in manual occupations, who accumulated human capital at a high rate over the sample period. Under the stable distribution assumption, the wage growth generated by this accumulation would instead be attributed to a growing relative price of manual tasks, consistent with the findings of the previous literature.

To provide a foundation for the large movements in human capital, particularly for manual workers, Section 3.3 explores the full path of human capital by cohort and age, separating changes in initial human capital at entry from changes in human capital by age. It shows that the growth is concentrated in manual occupations because it works mainly through the second channel: successive, increasingly educated cohorts follow steeper accumulation paths, and a counterfactual that holds each occupation's educational composition fixed at its 1970 level attributes nearly all of the manual growth to this upskilling.

3.2 Implications for Routine-Biased Technical Change

Having established the evolution of prices and quantities of occupational human capital, I now use these series to examine how each component contributes to the observed wage premium between occupational groups and to assess which predicted effects of technical change on task prices are supported by the data.

One of the active debates in the literature has been the role of routine-biased technical change (RBTC) in explaining changes in wage inequality. A key implication of RBTC, which Böhm (2020) notes has received relatively little attention until recently, is that task prices should polarize, with both abstract and manual task prices rising relative to routine tasks (Autor, Katz, and Kearney 2006; Acemoglu and Autor 2011; Cortes 2016; Böhm 2020). As the cost of computing capital falls, firms increasingly substitute capital for workers performing routine tasks, reducing the price of routine task input. At the same time, cheaper routine task production increases the productivity of workers performing abstract and manual tasks that complement routine activities, raising the relative prices of both types of nonroutine tasks.

Importantly, the RBTC literature has primarily focused on how technical change affects

wages through changes in task prices rather than through changes in the quantity of human capital supplied by workers to different tasks. However, the broader literature on skill-biased technical change (SBTC) has long emphasized that the relative price of skills depends on the race between skill-biased demand growth and the supply of skills (Tinbergen 1974; Katz and Murphy 1992; Acemoglu and Autor 2011).[21] The quantity channel documented in this paper, which operates through inter-cohort changes in the level and accumulation profile of task-specific human capital, has received relatively little attention in the existing RBTC literature.

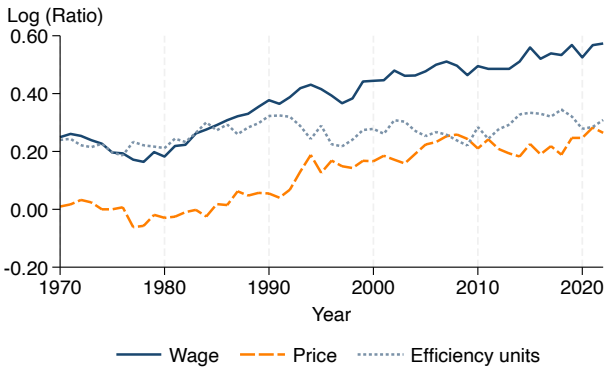
Figure 5 plots the annual series underlying the relative changes reported in columns (4) and (5) of Table 1. Panel (a) shows the abstract wage premium widening alongside the relative price of abstract human capital since the late 1970s, while panel (b) shows the manual wage premium improving against a declining relative price of manual human capital. The decomposition is therefore a feature of the series throughout the sample and not only of their 1970–2022 endpoints.[22]

Taken together, my findings support the role of routine-biased technical change in increasing the relative price of abstract tasks and in driving the widening of wage inequality at the top of the wage distribution.[23] However, they imply that the compression of wage

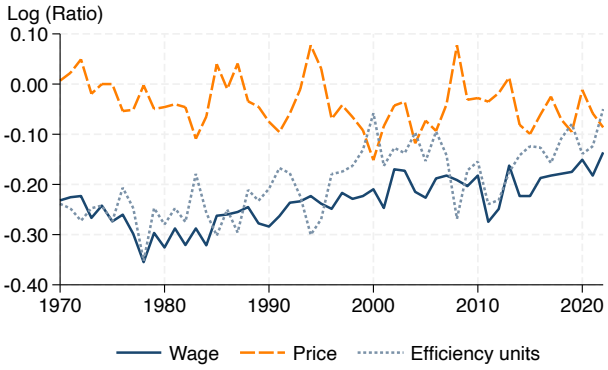
[21] In the canonical model of SBTC, this mechanism operates through aggregate skill quantities rather than through the quantity of human capital supplied to specific tasks. Skills and tasks are effectively treated as synonymous: high-skill workers perform high-skill tasks and low-skill workers perform low-skill tasks by assumption, imposing a one-to-one mapping between skills and tasks (Acemoglu and Autor 2011). As a result, the distinction between a worker's skill endowment and the tasks to which that endowment is applied, which is central to the RBTC framework, plays no role in the canonical SBTC model.

[22] A recent literature documents a sharp compression in the lower tail of the U.S. wage distribution following the Covid-19 pandemic (Autor, Dube, and McGrew 2023). This compression is not clearly visible as a comparable break in the between-occupation series estimated here. The manual-relative-to-routine wage gap narrows modestly at the end of the sample but stays within the range of its historical fluctuation, and the decomposition attributes this narrowing to growth in the relative quantity of manual human capital rather than to a rise in the relative price of manual occupations. The competition-based mechanism emphasized in that work operates through reductions in employer wage markdowns and between-firm reallocation, a margin that the flat spot method does not separately identify, and the price estimates for 2020–2022 are likely affected by the pandemic recession. Restricting the sample to 1970–2019 leaves the price of manual human capital declining by 23.0 log points (against 25.9 over the full sample) and its quantity growing by 13.5 log points (against 16.5), so the measured manual quantity growth is overwhelmingly secular rather than a product of the post-2020 compression.

[23] A growing literature highlights the increasing importance of social skills and documents rising returns to social tasks over this period (Deming 2017b; Edin et al. 2022). Because abstract occupations are often intensive in social tasks, the increase in the abstract price could reflect a rising return to social skills rather than to abstract task content. Appendix L shows that when occupations are instead grouped by social-task intensity, prices decline for both high- and low-social-task groups, because the high-social-task group pools



(a) Abstract Relative to Routine



(b) Manual Relative to Routine

Figure 5: Decomposition of Wage Premium

Notes: This figure shows the evolution of the estimated log relative prices, log relative hourly wages, and log relative efficiency units of human capital in the abstract and manual occupational groups relative to the routine occupational group. The sample includes full-time, full-year male wage and salary workers aged 25–64 years in the MCPS for the log relative hourly wages and log relative efficiency units of human capital series. For the log relative price series, the sample includes full-time, full-year male wage and salary workers in their flat spot age range in the MCPS. The estimation uses median hourly wages computed with CPS sampling weights.

inequality at the bottom was driven by a different force: the increase in the human capital stock in manual occupations. I do not find the rise in relative manual task prices that routine-biased technical change predicts under the assumption of an unchanged supply of human capital. While this pattern does not rule out the presence of routine-biased technical change in manual tasks, it suggests that any increase in the relative demand for manual tasks was largely offset by an expanding supply of human capital in those occupations. This interpretation is consistent with the broader emphasis in the skill-biased technical change

abstract workers with routine and manual ones. The social taxonomy therefore masks the absolute price gains the baseline classification identifies, indicating that the abstract price increase over the sample period is tied to abstract task content rather than to social-skill intensity.

literature on the race between relative demand and relative supply of skills (Tinbergen 1974; Katz and Murphy 1992; Goldin and Katz 2008), a mechanism that operates independently of whether the underlying demand shift is driven by routine-biased or other forms of technical change.

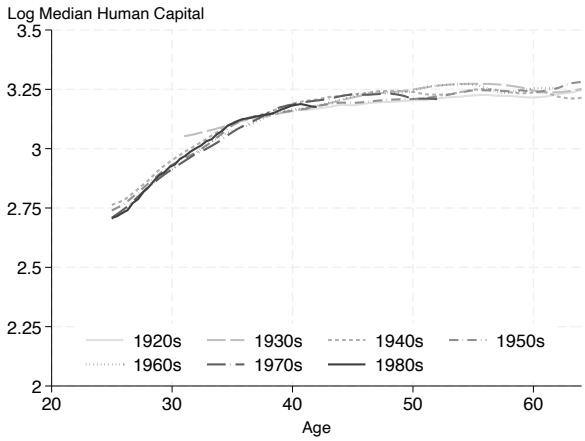
Additionally, my findings help explain the decline in human capital prices for college graduates documented by Bowlus and Robinson (2012). As college graduates increasingly sort into routine and manual occupations, the price of their human capital is differentially affected by the task content of their occupation. I find that while the price per efficiency unit of human capital for college-educated workers in abstract occupations has remained stable since the mid-1990s, it has steadily declined for college-educated workers in routine occupations (see Appendix K). These results highlight the significant role of occupational heterogeneity in explaining the evolving dynamics of wage inequality among college graduates, and provide further evidence that the value of a college education depends on the task content of workers' occupations.

3.3 Inter-Cohort Differences in Human Capital Accumulation

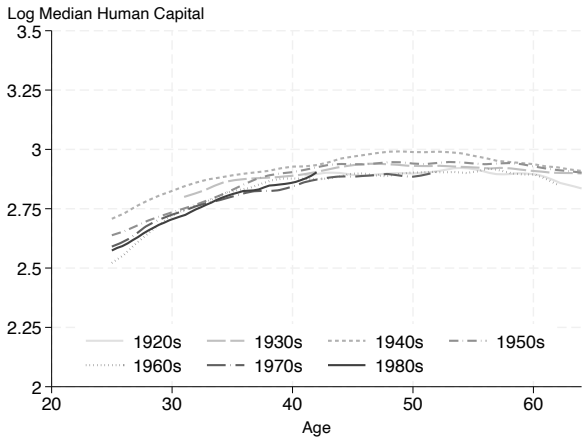
The price and quantity series estimated in Subsection 3.1 show that the quantity of human capital supplied to manual occupations grew substantially over the sample period, whereas growth in other occupations remained modest. To understand what drives these patterns, I first examine when over the life cycle this growth occurs by constructing cohort profiles of human capital by age. These profiles distinguish two sources of human capital growth: inter-cohort differences in human capital levels at age 25, an approximate age of labor market entry, and differences across cohorts in rates of human capital accumulation over the life cycle. The former reflects cohort-level differences in the human capital endowments that junior workers bring to occupations, while the latter reflects differences across cohorts in either on-the-job learning or dynamic selection into occupational employment over the life cycle.

Figure 6 depicts the cohort profiles of log median efficiency units of human capital by age for workers in different occupations. To construct each curve, I first compute individual-level efficiency units as the ratio of hourly wages to the estimated price of human capital, compute the log median across workers within each 10-year birth cohort and age cell, retaining

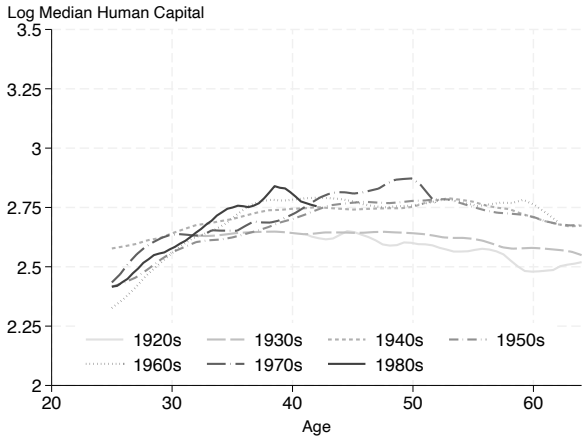
only cells with at least 40 observations and cohorts with at least 15 observed age points. I then perform a nonparametric kernel regression of these log median efficiency units on age separately for each cohort, where the Epanechnikov kernel function and rule-of-thumb bandwidth estimator are applied. The smoothed values are displayed on the graph.



(a) Abstract



(b) Routine



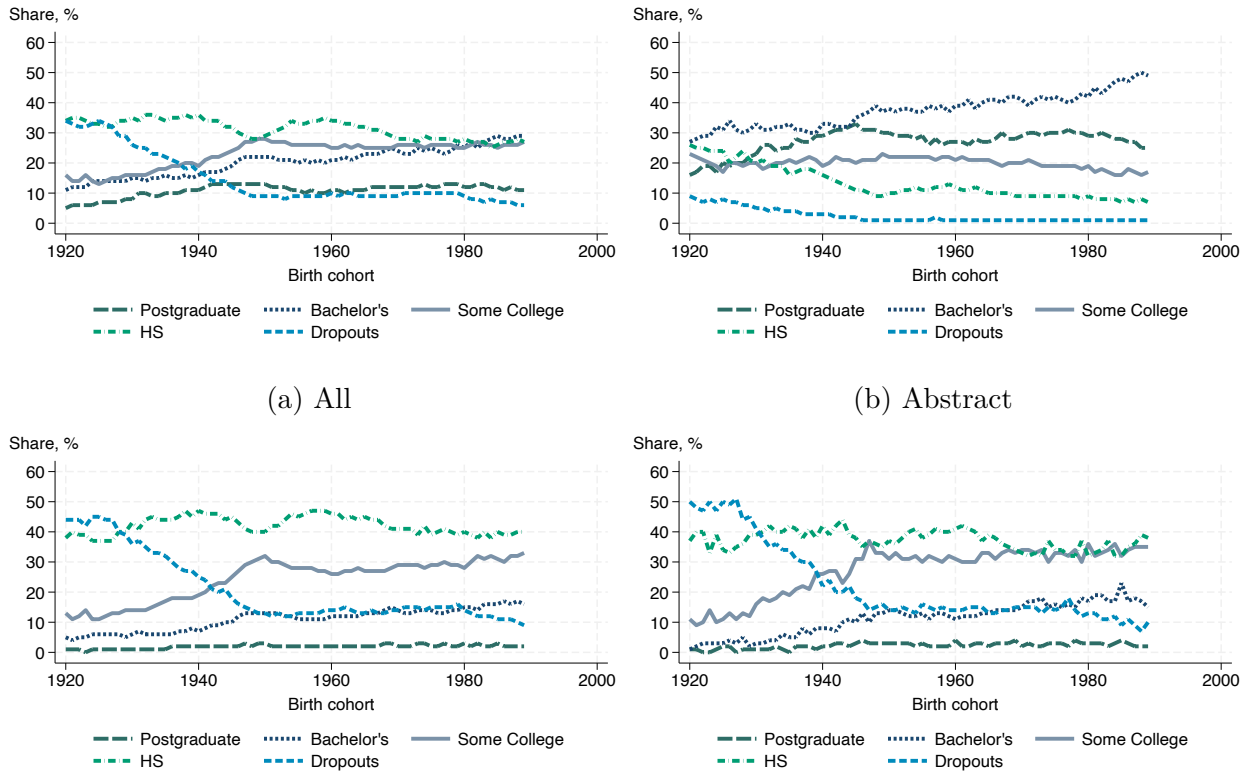
(c) Manual

Figure 6: Cohort Profiles of Log Median Efficiency Units of Human Capital

Notes: This figure shows the birth cohort profiles of log median efficiency units of human capital in abstract, routine, and manual occupational groups for full-time, full-year male wage and salary workers in the MCPS. Individual-level efficiency units are computed as the ratio of hourly wages to the estimated price of human capital. Log median efficiency units are computed within each 10-year birth cohort and age cell, retaining only cells with at least 40 observations and cohorts with at least 15 observed age points. Each curve is obtained by nonparametric kernel regression of log median efficiency units on age, using an Epanechnikov kernel and rule-of-thumb bandwidth, estimated separately for each cohort. The estimation uses median hourly wages computed with CPS sampling weights.

Consistent with the limited change in the quantity of human capital supplied to abstract occupations documented in Subsection 3.1, workers in abstract occupations experienced almost no inter-cohort differences in human capital, with cohort profiles overlapping and lying very close to one another across the entire age range. Cohort profiles for workers in routine occupations also lie close to each other, with the notable exception of the 1940s cohort, which lies somewhat above the others in mid-career. Aside from this shift, the profiles show similar shapes and accumulation rates across cohorts, indicating limited inter-cohort variation in human capital for routine workers. By contrast, human capital profiles for workers in manual occupations vary substantially across cohorts. Earlier cohorts (1920s through 1930s) show mildly declining life-cycle profiles at progressively higher levels. For later cohorts, particularly those born in the 1950s through 1980s, the profiles become distinctly steeper in early and mid-career, indicating faster human capital growth with age for younger workers. Returning to the two sources of growth, these patterns point to faster accumulation over the life cycle in more recent cohorts, rather than to higher human capital at age 25, as the main driver of the growth in manual occupations.

Several forces could drive this faster accumulation, including inflows of higher-ability workers from other occupations, improvements in population health, and educational upskilling, the rise over time in the average education of workers entering manual occupations. Among these, educational upskilling is the most directly testable with the available data. To assess whether it can account for these cohort differences, Figure 7 first documents how the educational composition of workers in each occupation evolved across birth cohorts. Across all occupations, successive cohorts are increasingly drawn from college-educated workers, reflecting the secular expansion of higher education. In abstract occupations this composition is already highly educated and changes little across cohorts, mirroring the overlapping cohort profiles of human capital. In manual and routine occupations, later cohorts are increasingly composed of workers with some college or a college degree, while the shares of high-school dropouts and graduates decline. This upskilling is most pronounced in manual occupations, where among the most recent cohorts workers with at least some college education reached a share comparable to that of high school graduates, whereas in routine occupations high



(a) All

(b) Abstract

(c) Routine

(d) Manual

Figure 7: Educational Composition by Birth Cohort

Notes: This figure summarizes the educational composition of workers across birth cohorts, for all occupations combined in panel (a) and within abstract, routine, and manual occupational groups in panels (b)–(d). The sample consists of full-time, full-year male wage and salary workers aged 25–64 years in the MCPS, and excludes self-employed workers and workers without records of occupation, annual earnings, and annual hours worked. The figure is restricted to birth cohorts [1920–1989], the range for which life-cycle human capital profiles are presented in Figure 6. All shares are computed using CPS sampling weights.

school graduates remained the largest group.[24]

To quantify the contribution of educational upskilling, Figure 8 compares the growth

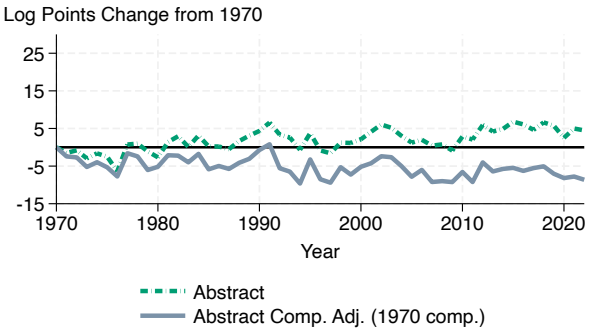
[24]This compositional shift has implications for the flat spot assumption. Consistent with the previous literature, I hold the flat spot assumption fixed over the sample period (Heckman et al. 1998; Bowlus and Robinson 2012; Lagakos et al. 2018; Fang and Qiu 2023). However, Bowlus and Robinson (2012) assign later flat spot ranges to more educated workers, implying that the educational upskilling documented here would shift the appropriate flat spot for manual workers toward later ages over time. A fixed range of 46–55 therefore becomes increasingly early relative to the true flat spot. Because a flat spot set too early absorbs wage growth generated by continued human capital accumulation, it biases the estimated price change upward and, as a residual, the estimated quantity change downward. Consistent with this direction of bias, Table 1 shows that using a later manual flat spot of 48–57 yields a larger price decline and greater quantity growth than under the baseline specification. If anything, the fixed flat spot assumption therefore causes the baseline estimates to understate both the decline in the price and the growth in the quantity of manual human capital.

rate of median human capital quantity against a composition-adjusted series that holds the educational composition fixed at 1970 levels. The composition-adjusted series is constructed by computing median human capital quantity separately for each of four education groups within each occupation and year, less than high school, high school diploma, some college, and college or above, where workers with bachelor's and postgraduate degrees are pooled due to the small number of postgraduate workers in manual occupations early in the sample, and then aggregating across groups using their within-occupation employment shares in 1970 as fixed weights. The reweighting exercise suggests that educational composition explains all of the human capital growth observed in manual occupations. By 2022, observed median human capital in manual occupations stands 16.5 log points above its 1970 level, while the composition-adjusted series stands 3.6 log points below it, so educational upgrading contributes 20.1 log points. The role of educational upskilling is also evident across detailed occupations within the manual group, where occupations with larger increases in educational attainment, such as childcare workers and firefighters, also show greater growth in human capital quantity (see Appendix Figure M.2).

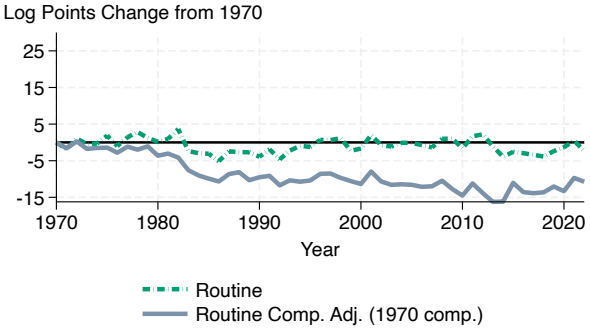
Educational upskilling operates in all occupational groups, with the composition-adjusted series lying below the observed one in each of them.[25] What distinguishes manual occupations is the size of both components. Routine occupations provide the natural comparison because routine and manual workers have similar educational attainment throughout the sample (see Appendix Figure F.1) and experienced broadly similar patterns of educational upskilling (see Figure 2, panels (b) and (c)). By 2022, upskilling contributes 8.4 log points in routine occupations, less than half the manual figure, while the composition-adjusted series lies 10.8 log points below its 1970 level, against 3.6 log points in manual occupations. Observed human capital in routine occupations is accordingly 2.4 log points below its 1970 level, so educational upskilling offsets most of the underlying decline there rather than generating growth. Since the two groups experienced similar educational upskilling, their divergent outcomes suggest that education translates into human capital accumulation differently across occupations.

[25]The reweighting holds educational composition fixed but not the composition of workers within education groups or the age composition of each occupation, so the adjusted series reflects both channels and this exercise is not designed to separate them. The contribution of educational upskilling is the difference between two series deflated by the same occupation-specific price and is therefore unaffected by the estimated price series.

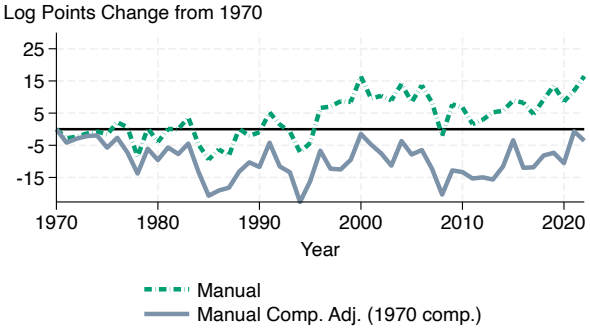
One reason this may occur is that education affects subsequent human capital accumulation through the experience profile. Lagakos, Moll, Porzio, Qian, and Schoellman (2018) show that experience–human capital profiles are substantially steeper for college-educated workers, and Bowlus and Robinson (2012) document substantial inter-cohort human capital growth for college-educated workers despite similar price changes across education groups. The same gradient appears within manual occupations in my sample: workers with at least some college education accumulate human capital faster early in the life cycle than those with a high school degree or less (see Appendix Figure M.1). As the share of manual entrants with at least some college education has risen, more of each cohort has followed this steeper path, raising the average profile in manual occupations. In routine occupations, this gradient is weaker because workers with at least some college education accumulate human capital only slightly faster than those with at most a high-school diploma. As a result, even as routine occupations upskill, their cohort profiles steepen less than those of manual workers. Intuitively, schooling keeps paying off in human capital accumulation only where a job leaves something to learn. Routine tasks can be reduced to explicit, codifiable rules (Autor, Levy, and Murnane 2003; Acemoglu and Autor 2011) that require little problem-solving once learned, whereas non-routine service tasks, such as those of a detective or childcare worker, continuously reward judgment and problem-solving that more-educated workers keep developing with experience.



(a) Abstract



(b) Routine



(c) Manual

Figure 8: Observed and composition-adjusted growth in human capital relative to 1970

Notes: This figure displays the observed and composition-adjusted growth in median human capital quantity in occupations relative to 1970, where human capital quantity is measured as the median hourly wage deflated by the estimated occupation-specific price. The composition-adjusted series is constructed by reweighting median human capital quantity across education-occupation-year cells using the education shares of each occupation in 1970. Education groups are less than high school, high school diploma, some college, and college or above, where workers with bachelor's and postgraduate degrees are pooled. The sample includes full-time, full-year male wage and salary workers in the MCPS 1971–2023. Median hourly wages, means and shares are computed using CPS sampling weights.

3.4 The Evolution of Human Capital for Female Workers

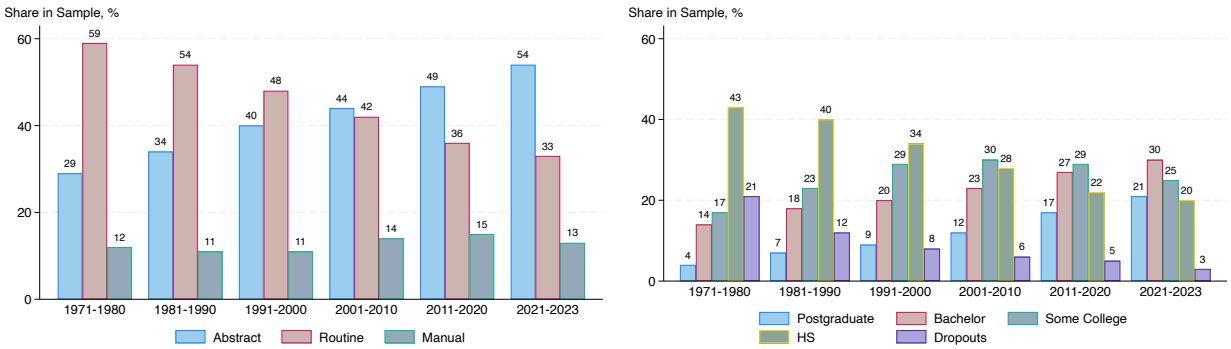
One of the major changes in labor markets over the last half century has been the dramatic increase in the attachment of female workers to the labor market (Goldin 2006). As women became a substantial part of the labor force, the incentives for female workers to invest in their human capital through work experience or education have also increased (Bowlus and Robinson 2020). Since female workers experienced larger shifts in both occupational and educational composition compared to male workers over the sample period, cohort effects in human capital may be particularly pronounced for women.

Figure 9 depicts the changing composition of full-time, full-year female workers in the MCPS. Panel (a) shows that the occupational composition of female workers shifted more dramatically than that of male workers. The share of female workers in routine occupations fell from 59% in the 1970s to 36% in the 2010s, over twice the decline observed for male workers over the same period. This contraction was accompanied by a substantial expansion of abstract employment, from 29% to 54% by the early 2020s, driven in part by the rapid growth of college education among women. The share of female workers with bachelor's or postgraduate degrees rose from 18% in the 1970s to 51% in the 2020s. Female workers also consistently had a higher presence in manual occupations than their male counterparts, and this share expanded further over the sample period.

To examine how these trends are reflected in human capital accumulation, I apply the male-estimated price series to female wages to recover human capital quantity series for female workers in each occupational group.[26] This approach assumes that male and female workers within the same occupational group face the same price per efficiency unit of human capital, and that an efficiency unit of male human capital of a given type is a perfect substitute for an efficiency unit of female human capital of the same type. Under standard competitive market conditions, this implies that both groups face the same market price for their occupational human capital, while allowing the quantity of human capital to differ across sexes.[27]

[26]This approach is similar to Bowlus and Robinson (2020), who apply male-estimated education-based price series to quantify changes in the quantity of female human capital, and to Yamaguchi (2018), who impose no gender differences in the returns to task-specific skills.

[27]Estimating a separate price series for female human capital using cross-sectional data is complicated



(a) Occupation (b) Education

Figure 9: Female Sample Composition by Decade

Notes: This figure shows summary characteristics for the sample of full-time, full-year female wage and salary workers aged 25–64 years in the MCPS 1971–2023. The sample excludes self-employed workers and workers without records of occupation, annual earnings, and annual hours worked. Panel (a) summarizes the sample's occupational composition, while panel (b) summarizes its educational composition. Means and shares are computed using CPS sampling weights.

This common-price assumption is not innocuous. A large literature documents a persistent gender wage gap of which human capital explains little and occupation and industry a substantial share, with a residual remaining after these controls, potentially driven by discrimination (Blau and Kahn 2017). Whether discrimination biases the estimates depends on the channel through which it operates. If it works through barriers to accumulation, such as restricted access to training and promotion, it lowers the human capital women acquire rather than the price they receive, and the estimates of inter-cohort changes remain valid however these barriers evolved. If instead it introduces a wedge, paying women less for the same occupational human capital, then dividing their wages by the male price series understates their human capital. A constant wedge leaves the estimated cohort and time variation unaffected; only a secular narrowing biases the estimated growth upward. The cohort and time variation should therefore be read as that of realized female occupational human capital, with the estimated growth an upper bound on the truth to the extent the wedge narrowed.[28]

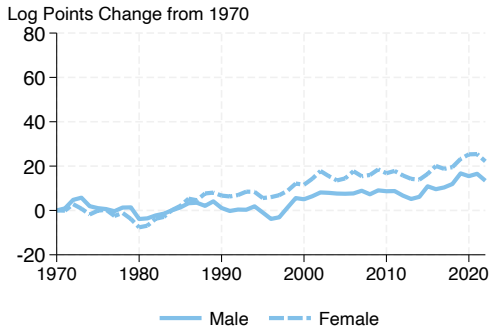
by substantial differences in how closely age approximates actual experience for male and female workers. For female workers, this approximation depends on a variety of factors, including fertility, marriage, and part-time work, and has changed over time as successive cohorts of women remained in the labor force for longer periods over the life cycle.

[28]The residual in an Oaxaca-Blinder decomposition that survives controls for human capital, occupation,

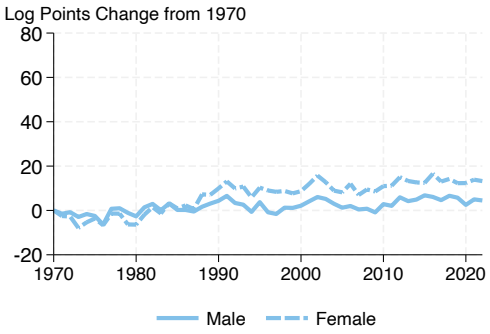
Across all occupational groups, female workers have experienced larger growth in median hourly wages and human capital accumulation than male workers. Figure 10 contrasts the log points change in median hourly wages and efficiency units of human capital for female and male workers relative to 1970.[29] While both male and female workers in abstract occupations experienced growth in hourly wages and human capital levels, this growth was larger for female workers. In routine occupations, male workers experienced a decline in hourly wages while their human capital levels remained relatively stable; by contrast, both hourly wages and human capital levels of female workers increased over the sample period. Finally, female workers in manual occupations experienced a dramatic increase in human capital quantities across cohorts, which, despite the simultaneous decline in the price of manual occupational human capital, translated into substantial growth in hourly wages.

and industry is the closest analog to such a wedge, and Blau and Kahn (2017) find that it fell substantially during the 1980s but changed little afterward. A narrowing wedge is therefore likely to be most relevant for wage observations from the 1980s and earlier, leaving more recent cohorts largely unaffected.

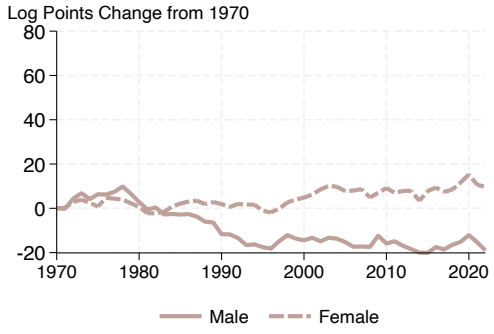
[29]Efficiency units of human capital are computed as the ratio of median hourly wages to the price of human capital estimated from the male sample.



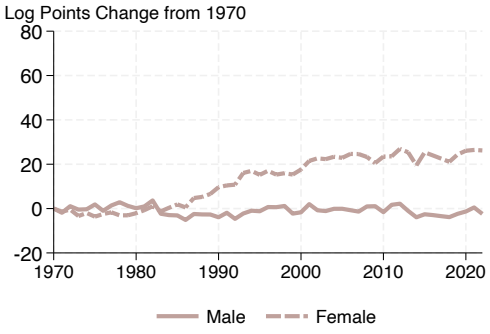
(a) Abstract: Log Median Hourly Wage



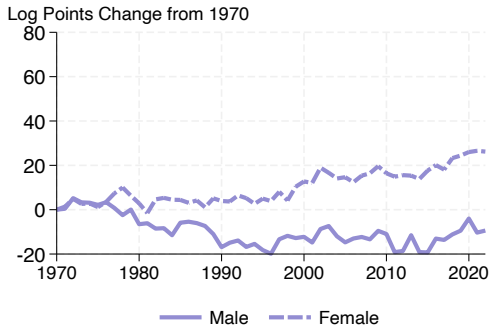
(b) Abstract: Log Efficiency Units



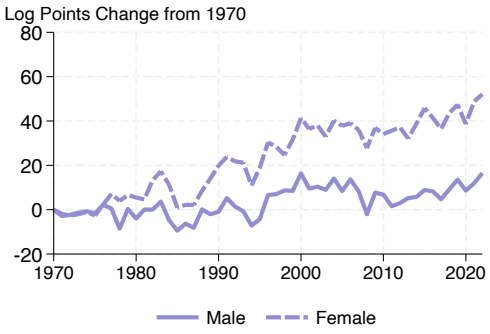
(c) Routine: Log Median Hourly Wage



(d) Routine: Log Efficiency Units



(e) Manual: Log Median Hourly Wage



(f) Manual: Log Efficiency Units

Figure 10: The Change in Hourly Wages and Quantities of Occupational Human Capital: Male and Female Workers

Notes: This figure shows the evolution of log median hourly wages and log quantities of human capital in abstract, routine, and manual occupational groups for male and female full-time, full-year wage and salary workers in the MCPS. All series are normalized to zero in 1970 and expressed as log point changes relative to 1970. Solid lines denote male workers; dashed lines denote female workers. Log efficiency units are computed as the difference between log median hourly wages and the log price of human capital, where the price series are estimated using male workers in their flat spot age range. The estimation uses median hourly wages computed with CPS sampling weights.

4 Conclusion

I examine the sources of the growing wage inequality between workers in abstract, routine, and manual occupations by allowing for across-cohort changes in the distribution of human capital. Applying the flat spot price identification method developed in Heckman et al. (1998) and Bowlus and Robinson (2012) to MCPS data from 1970 to 2022, I estimate the evolution of prices and quantities of human capital supplied to abstract, routine, and manual occupational groups.

The results indicate that the widening of wage inequality at the top and the compression of wage inequality at the bottom of the distribution are driven by different forces. The price of human capital in abstract occupations increased relative to both routine and manual occupations, consistent with the role of routine-biased technical change in raising the relative demand for abstract tasks. By contrast, the rising wage premium of manual relative to routine workers is driven not by an increase in the relative price of manual tasks, as routine-biased technical change would predict, but by faster inter-cohort growth in the quantity of human capital supplied to manual occupations. Decomposing this growth shows that educational upskilling, the rise over time in the share of college-educated workers entering manual occupations, accounts for all of it. This pattern is consistent with the race between relative demand and relative supply of skills emphasized in the skill-biased technical change literature, in which any rise in the relative demand for manual tasks was offset by an expanding supply of human capital in those occupations.

By focusing on workers' occupations rather than education, I explain the puzzling decrease in the price of human capital for college graduates documented in Bowlus and Robinson (2012) as a consequence of their increasing employment in routine and manual occupations, where the price of human capital has declined. Therefore, my findings provide evidence that technological change affects workers differently depending on the task content of their occupations, even among those with similar education levels.

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Appendix: The Evolution of Prices and Quantities of Occupational Human Capital

A Data: Imputing Consistent Annual Hours Worked in the MCPS

For surveys conducted after 1976, the MCPS reports the exact number of weeks the respondent worked for profit, pay, or as an unpaid family worker during the preceding calendar year. However, before 1976, this information was only provided in intervals. To estimate the number of weeks worked for years prior to 1976, I assign a value based on the mean number of weeks worked by wage and salaried workers aged 16 to 64 years with positive earnings and valid occupations within the corresponding intervals in 1976–2023.

Interval	Assigned Number of Weeks
1 - 13 weeks	8.2
14-26 weeks	21.5
27-39 weeks	33.5
40-47 weeks	42.5
48-49 weeks	48.3
50-52 weeks	51.9

Table A.1: Extrapolation of weeks worked last year prior to 1976 based on reported intervals.

Note: The assigned number of weeks corresponds to the average weeks worked within each interval for wage and salaried workers aged 16 to 64 with positive earnings and valid occupations, based on the MCPS data from 1976–2023.

For surveys conducted after 1976, the MCPS reports the number of hours per week that respondents *usually worked* if they were employed during the previous calendar year. However, for surveys prior to 1976, the MCPS only reports the total number of hours respondents worked *during the previous week*. To impute the usual hours per week last year for respondents in the pre-1976 surveys, I follow a two-step approach.

First, for workers who reported positive hours worked last week, their "hours last week" is used as an estimate for "usual hours per week last year." Second, for workers who did not report positive hours last week but worked last year, I predict their "usual hours per week last year" using a regression of "hours last week" on age, years of schooling, gender, and an occupational group indicator for each survey year from 1971 to 1975. The sample

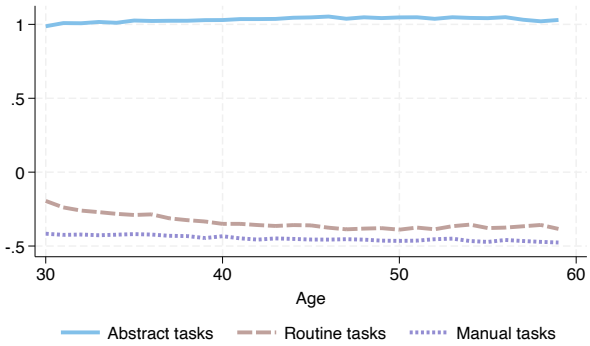
for regressions includes wage and salaried workers aged 16 to 64 years with positive earnings and valid occupations.

B Differences in Job Tasks Across Occupations

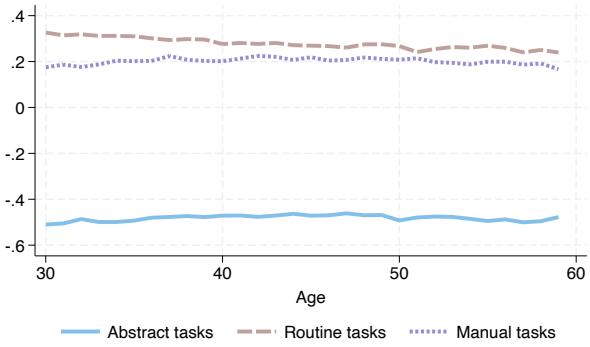
Following Acemoglu and Autor (2011), I sort workers into broadly defined occupational groups based on the Census classification of occupations. While commonly used in the literature (Beaudry, Green, and Sand 2016; Cortes 2016; Böhm 2020; Cavaglia and Etheridge 2020), it is possible that this grouping does not reflect workers' occupational tasks. This section shows that this occupational grouping preserves the relative ranking of occupational groups in terms of their task intensity.

To measure average task intensity in occupational groups, I use the crosswalk provided by Autor and Dorn (2013) to map the average of D.O.T. 1977 task variables to workers' occupations. I follow Autor and Dorn (2013) and construct a measure of abstract task intensity as the average of "direction, control, and planning of activities" and "quantitative reasoning requirements" scores. The measure of routine task intensity is the average of "adaptability to work requiring set limits, tolerances, or standards" and "finger dexterity" scores. The measure of manual task intensity is the "eye, hand, and foot coordination" score. I standardize all task intensity measures to have a zero mean and a standard deviation equal to one.

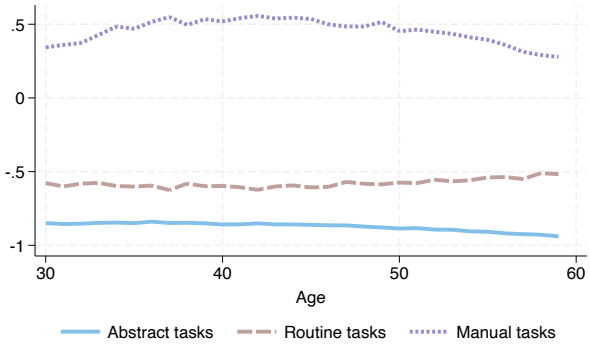
Figure B.1 shows the evolution of the average task intensity for all cohorts of workers by age. The average intensity of abstract tasks exceeds the average intensities of routine and manual tasks for workers in the abstract occupational group. Moreover, the intensity of abstract tasks remains stable across ages, implying that comparable occupational tasks, as measured at the occupation level, are assigned to workers of different ages. The average intensity of routine tasks is the highest in routine occupations compared to other occupations. Routine tasks also have the highest intensity in the routine group compared to abstract and manual tasks. Finally, manual task intensity is the highest for the manual group. While the average intensity of manual tasks exhibits a slight bell-curve shape, the average manual task intensity remains relatively stable for workers in their flat spot age 46–55.



(a) Abstract



(b) Routine



(c) Manual

Figure B.1: Average Task Intensity in Occupational Group by Age

Notes: This figure shows the life-cycle profile of the average intensity of abstract, routine, and manual tasks by broad occupational groups. The tasks are computed using D.O.T. 1977 task data provided in Autor and Dorn (2013). The sample includes FTFY wage and salaried male workers in the MCPS.

C Derivation of the Wage Decomposition

Individual-Level Identity

The period t hourly wage for worker i of age a in occupational group $o \in \{A, R, M\}$ is given by

$$W_{i,t}^{o,a} = P_t^o \times H_{i,t}^{o,a}, \tag{1}$$

where P_t^o is the price of one efficiency unit of human capital in occupation o in year t, and $H_{i,t}^{o,a}$ is the number of efficiency units supplied by worker i. Taking logarithms yields

$$\ln W_{i,t}^{o,a} = \ln P_t^o + \ln H_{i,t}^{o,a}. \tag{2}$$

This is an exact accounting identity at the individual level. Throughout the rest of the appendix, I use lower-case letters to denote log-transformed variables, so that $w_{i,t}^{o,a} \equiv \ln W_{i,t}^{o,a}$, $p_t^o = \ln P_t^o$ and $h_{i,t}^{o,a} \equiv \ln H_{i,t}^{o,a}$. Equation (2) can then be written compactly as

$$w_{i,t}^{o,a} = p_t^o + h_{i,t}^{o,a}. \tag{3}$$

Aggregation via Medians

At the group level, I work with the median log wage of workers of age a in occupation o in year t. Since P_t^o is common to all workers in occupation o in year t, that is, it does not vary across individuals i, it passes through the median operator exactly:

$$\mathrm{Med}_i\big[w_{i,t}^{o,a}\big] \;=\; p_t^o + \mathrm{Med}_i\big[h_{i,t}^{o,a}\big]. \tag{4}$$

The accounting identity is therefore preserved exactly under the median. No approximation is involved; the only requirement is that P_t^o be common across workers within an occupation-year cell, which is a maintained assumption of the model.

Identification of Price Changes via the Flat Spot

The flat spot assumption states that workers in the flat spot age range a^* have stable human capital levels across consecutive years within the same synthetic cohort:

$$E\big[h_{i,t}^{o,a^*} - h_{i,t-1}^{o,a^*-1}\big] = 0. \tag{5}$$

This implies that the mean of $h_{i,t}^{o,a^*}$ and the mean of $h_{i,t-1}^{o,a^*-1}$ are equal across consecutive years within the same synthetic cohort. To move from equality of means to equality of medians, I additionally assume that the distribution of human capital for a synthetic cohort is stable within the flat spot age range, so that

$$\text{Med}_i\big[h_{i,t}^{o,a^*}\big] = \text{Med}_i\big[h_{i,t-1}^{o,a^*-1}\big]. \tag{6}$$

In practice, for each consecutive pair of years $t-1$ and t, the MCPS provides a representative sample of workers aged a^* in year t and workers aged a^*-1 in year $t-1$. Abstracting from mortality, these form a synthetic cohort. Substituting (6) into the difference of (4) across consecutive years gives

$$\text{Med}_i\big[w_{i,t}^{o,a^*}\big] - \text{Med}_i\big[w_{i,t-1}^{o,a^*-1}\big] \;=\; p_t^o - p_{t-1}^o. \tag{7}$$

The estimated log price change between $t-1$ and t is the average of these differences across all ages in the flat spot range. Therefore, with a flat spot window of length M years, the estimated log price change between $t-1$ and t is averaged across all $M-1$ cohort comparisons available within the flat spot window. Under the flat spot assumption, each of these comparisons provides an independent estimate of the same price change $\ln P_t^o - \ln P_{t-1}^o$. Averaging across all $M-1$ comparisons therefore reduces sampling error without introducing bias.

Decomposition of the Relative Wage

For any two occupational groups o and o', taking the difference of (4) across groups yields an exact additive decomposition of the log relative wage into a relative price component and a relative quantity component:

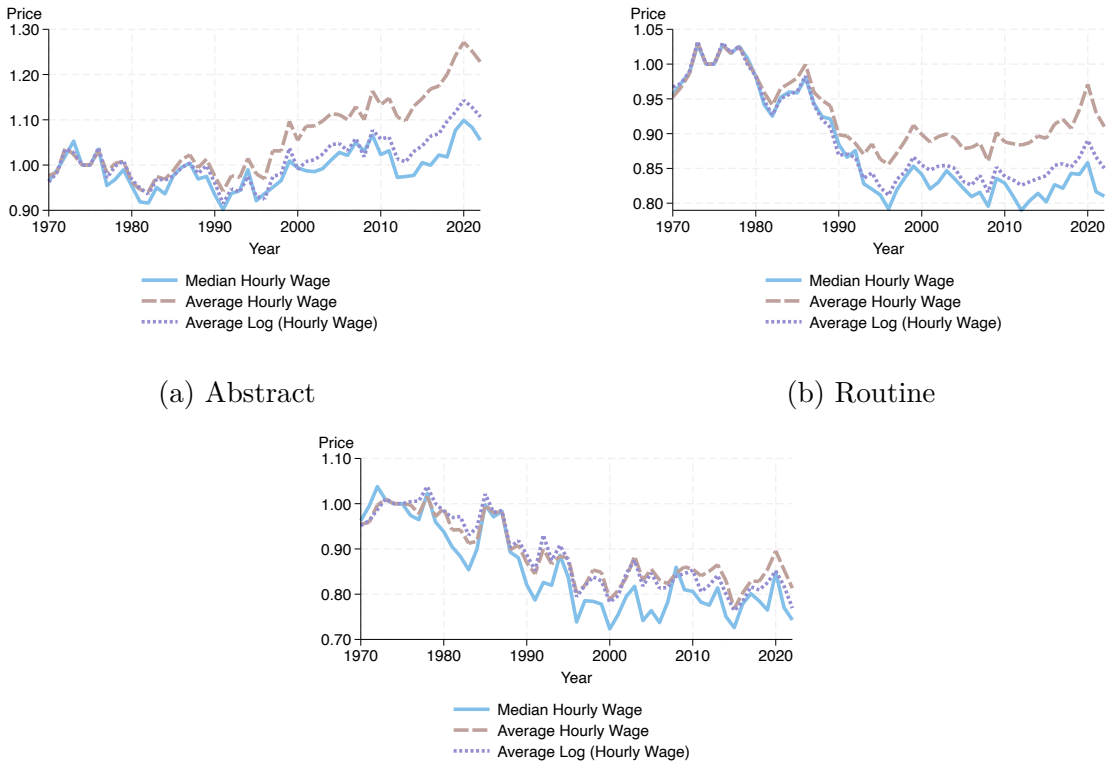
$$\bar{w}_t^{\,o} - \bar{w}_t^{\,o'} = \left(p_t^o - p_t^{o'}\right) + \left(\bar{h}_t^{\,o} - \bar{h}_t^{\,o'}\right), \tag{8}$$

where $\bar{w}_t^{\,o} \equiv \text{Med}_i[w_{i,t}^{o,a}]$ and $\bar{h}_t^{\,o} \equiv \text{Med}_i[h_{i,t}^{o,a}]$ denote the median log wage and median log efficiency units, respectively, for workers aged 25–64 in occupation o in year t, and $p_t^o \equiv \ln P_t^o$. The decomposition holds because p_t^o is occupation-specific and constant across individuals within each occupation-year cell, so it passes through the median operator exactly, and

taking differences across groups preserves the additive structure of (2).

D Sensitivity to Wage Measure

Using median hourly wages in the estimation allows me to avoid the issue of inconsistent top-coding of income values in the CPS. However, it is common in the labor economics literature to rely on average wages and average log wages in the analysis. This section shows that my findings are robust to the use of alternative wage measures. Figure D.1 compares the price series estimated using the median hourly wages, average hourly wages, and average log hourly wages. The increase in the price for abstract occupations estimated with the use of average wages is even larger than the one reported for the benchmark price series. The estimated fall in the price for routine occupations is lower in magnitude for the average wages than for the median wages. However, the direction of the change is preserved.



(a) Abstract

(b) Routine

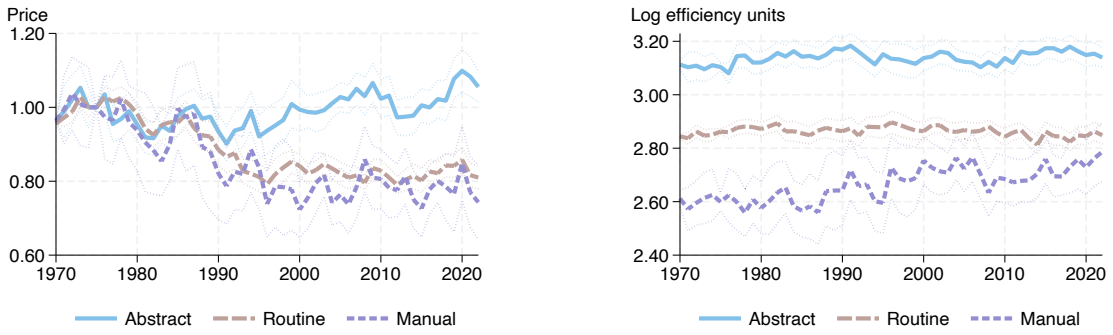
(c) Manual

Figure D.1: Sensitivity of Price Series to Using Alternative Wage Measures

Notes: This figure shows the evolution of estimated prices of human capital using average hourly wages, median hourly wages, and average log hourly wages to compute the price series. The sample includes full-time, full-year male wage and salary workers in their flat spot age ranges in the MCPS 1971–2023.

E Confidence Intervals for Annual Changes

I estimate median hourly wages, human capital price, and quantity series in each occupational group using quantile regressions. First, the log price change between periods $t-1$ and t is estimated as the increase in the log median wage between workers of age $a-1$ in period $t-1$ and workers of age a in period t, averaged across all workers in the flat spot age range. Second, the wage growth rate for a synthetic cohort of workers in period t is computed as the change between the log median wage of workers in year t averaged across all ages and the log median wage of workers in year $t-1$ averaged across all ages. Finally, the change in the log efficiency units of human capital is computed as the difference between the change in the log median wages and the log prices of human capital.



(a) Price Series

(b) Log Efficiency Unit Series

Figure E.1: Confidence Intervals: Prices and Quantities of Occupational Human Capital

Notes: This figure shows the evolution of estimated prices and quantities of human capital in abstract, routine, and manual occupational groups in the sample of full-time, full-year male wage and salary workers in the MCPS. The figure displays 95% confidence intervals for the estimated change in prices and log efficiency units of human capital between consecutive years. The bands represent uncertainty in each annual change, holding the preceding year's estimated level fixed.

Standard errors for annual log changes in prices, efficiency units, and relative prices are computed using the delta method via Stata's `nlcom` command, which accounts for the nonlinear transformation of quantile regression coefficients. These standard errors reflect the precision of each year's price change estimate. The cumulative price level in year t is obtained by summing all annual log price changes from 1970 to t. While I report point estimates for cumulative price changes below, reported confidence intervals represent the uncertainty in the annual price changes for each year, not the cumulative uncertainty in price levels.

F Flat Spot Identification

Given the age-time-cohort collinearity, additional assumptions are needed to determine a useful range of potential "flat spot" ages for workers in different occupations. To establish the appropriate flat spot age range for each occupation, I exploit the theoretical implications of life-cycle human capital investment models, observable features of cross-sectional wage profiles, and systematic differences in educational composition across occupations.

To understand how cohort effects manifest in observable wage patterns, it is useful to consider what determines the slope of a cross-sectional age-earnings profile. In any given year t, we observe workers of different ages, who belong to different birth cohorts. Let

$w_t^{o,a} \equiv E[w_{i,t}^{o,a}]$ denote the expected log wage of a worker of age a in occupational group o in year t. Following Fang and Qiu (2023), it can be written as:

$$w_t^{o,a} = p_t^o + h_t^{o,a} = p_t^o + s_{t-a}^o + r_a^o, \tag{9}$$

where $h_t^{o,a} = s_{t-a}^o + r_a^o$ decomposes log human capital into s_{t-a}^o, the log human capital at labor market entry for the cohort born in $t-a$ (cohort effect), and r_a^o, the return to age a (experience effect). The term p_t^o captures the log human capital price (time effect). Taking the derivative with respect to age, the slope of the cross-sectional profile is:

$$\frac{\partial w_t^{o,a}}{\partial a} = \dot{r}_a^o - \dot{s}_{t-a}^o, \tag{10}$$

where $\dot{r}_a^o \equiv \frac{\partial r_a^o}{\partial a}$ represents the rate of life-cycle human capital growth (experience effect) and $\dot{s}_{t-a}^o \equiv \frac{\partial s_c^o}{\partial c}\Big|_{c=t-a}$ represents the rate of intercohort human capital growth (cohort effect). Since both r^o and s^o are in logarithms, $\dot{r}^o$ and $\dot{s}^o$ are interpreted as growth rates. This decomposition reveals that the slope of the cross-sectional profile reflects a race between two forces. The experience effect $\dot{r}_a^o$ captures how human capital grows as workers age and tends to make the profile upward-sloping. The cohort effect $\dot{s}_c^o$ captures how human capital at the entry to the labor market differs across successive cohorts. If younger cohorts are more productive upon entry (positive cohort effect $\dot{s}_c^o > 0$), this force tends to flatten or invert the profile. The profile slopes upward when $\dot{r}_a^o > \dot{s}_{t-a}^o$, slopes downward when $\dot{r}_a^o < \dot{s}_{t-a}^o$, and peaks where these two forces balance.

This relationship can inform us about the potential flat spot age range if we can determine the direction of intercohort human capital growth by examining observable patterns in occupational shares across cohorts and changes in the educational composition of the workforce. With positive cohort effects ($\dot{s}_{t-a}^o > 0$), Equation 10 implies that if workers of adjacent ages experience cross-sectional wage growth ($\frac{\partial w_t^{o,a}}{\partial a} > 0$), the experience effect must be positive and strong enough to offset the negative effect from intercohort changes ($\dot{r}_a^o > \dot{s}_{t-a}^o$). Therefore, during the ages when the cross-sectional profile slopes upward and even reaches its peak, human capital must still be growing. The flat spot of the human capital profile, where $\dot{r}_a^o = 0$, must occur at later ages compared to the peak of the cross-sectional wage profile. This reasoning allows me to use observable features of cross-sectional wage profiles

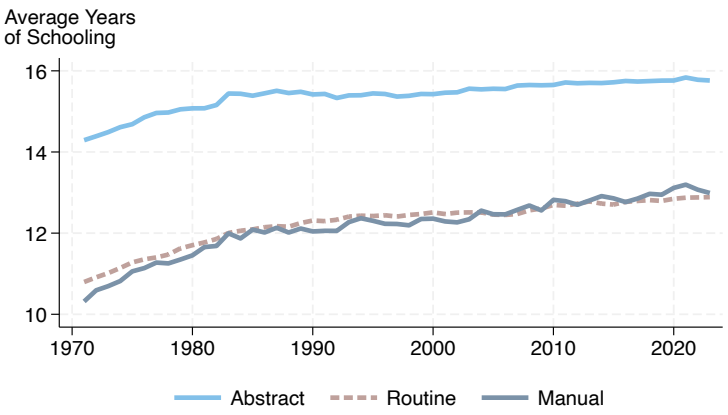


Figure F.1: The Growth of Average Schooling in Occupations

Notes: This figure shows the evolution of the average years of schooling in occupations. The sample includes full-time, full-year male wage and salary workers aged 25–64 years in the MCPS 1971–2023.

to place a lower bound on the flat spot age range.

I apply this reasoning separately to each occupational group, exploiting the hierarchical ranking of occupations by skill (Acemoglu and Autor 2011; Autor and Dorn 2013; Cortes 2016): abstract occupations constitute the highest-skill group, routine occupations the middle-skill group, and manual occupations the lowest-skill group. This ranking has direct implications for the direction of ability selection effects when occupational employment shares change across cohorts, as I discuss in turn below.

F.1 Flat Spot in Abstract Occupations

To predict the direction of intercohort changes in human capital in abstract occupations, I examine how ability selection and human capital production have evolved across cohorts. Abstract occupations consist primarily of college-educated workers, with workers switching into abstract jobs drawn disproportionately from the higher-wage deciles of routine and manual occupations, and the probability of switching to abstract jobs increasing in ability (Cortes 2016). If the initial ability distribution of the population is stable, a rising share of workers selecting into the highest-skill group implies that the marginal entrant has lower ability than the average incumbent. Therefore, expansion of the abstract occupation share within a birth cohort generates negative ability selection effects, while a declining share implies positive ability selection. At the same time, improvements in the human capital

production function enable workers to accumulate human capital at a higher rate for any given initial ability level, generating positive cohort effects. A declining share of abstract occupations thus operates through both channels simultaneously, unambiguously implying a positive cohort effect ($\dot{s}_c^A > 0$).

Panel (a) of Figure F.2 plots the share of full-time, full-year workers aged 35 to 40 in the MCPS who are employed in abstract occupations, who hold a bachelor's degree, and who hold a postgraduate degree by birth cohort.[30] For cohorts born in the early 1950s, the share of workers employed in abstract occupations declines, following a trajectory closely aligned with declines in bachelor's and postgraduate attainment. The parallel movement across occupational and educational measures points to cohort-level shifts in the composition of skills rather than changes specific to abstract-task demand.

These cohorts reached typical college-attendance ages during the late 1960s and 1970s, a period characterized by substantial institutional and social changes affecting educational pathways. In particular, the Vietnam War coincided with notable changes in post-secondary enrollment patterns, as military deferment policies linked schooling and service decisions (Card and Lemieux 2001b). The draft deferment for college enrollment boosted attendance rates during the war years for cohorts born in the late 1940s, while the end of the draft in 1973 removed this enrollment incentive for later cohorts. Additional factors contributing to the subsequent enrollment decline include cohort-size effects and changes in perceived returns to education (Card and Lemieux 2001a).

In 2001, the 1950 birth cohort reached age 51, while the 1956 birth cohort reached age 45. Cohorts born between 1950 and 1956 experienced declining shares of workers employed in abstract occupations. Within the framework of the model, this pattern is interpreted as reflecting a positive ability selection effect. In addition, Figure F.1 shows that the average level of schooling within abstract occupations increased over the period, consistent with positive shifts in human capital production functions (Bowlus and Robinson 2012). Taken together, these patterns imply a positive aggregate cohort effect for abstract workers born in the early 1950s ($\dot{s}_c^A > 0$).

Figure F.3 plots cross-sectional wage profiles in 2001 for workers in abstract occupations.

[30] Workers aged 35 to 40 are approaching their potential flat-spot age range.

The profile for abstract occupations peaks at age 57. Since the cross-sectional wage profile remains upwards-sloping at the age of 50, the positive cohort effect implies that the life-cycle human capital profile for abstract workers cannot peak earlier than age 50. Based on this reasoning, I set the baseline flat spot age range for abstract workers at ages 51–60, since setting the flat spot age range later than that may introduce bias due to non-random selection into retirement among older workers. This range begins one year later than the flat spot region identified for college-educated workers in Bowlus and Robinson (2012), reflecting the slightly older age at peak earnings observed in abstract occupations.

F.2 Flat Spot in Manual Occupations

I perform a similar analysis of cohort effects to determine a plausible flat spot age range for workers in manual occupations. Panel (b) of Figure F.2 plots the share of full-time, full-year workers aged 35 to 40 in the MCPS who are employed in manual occupations, who have graduated from or dropped out of high school. The share of workers in manual occupations has been steadily increasing. As manual occupations have the lowest median wages and are considered to be the lowest-skill occupations, expansion of employment in manual occupations implies that marginal workers with higher ability levels than the average worker in manual occupations are transferring to manual occupations, increasing the average levels of skill. These positive ability selection effects, as well as potential improvement from human capital production driven by the growing average educational attainment illustrated in Figure F.1, imply that manual occupations have been consistently experiencing the positive cohort effect ($\dot{s}_c^M > 0$).

Figure F.4 plots the cross-sectional median wage profiles within 10-year bins for workers in abstract, manual, and routine occupations. Panel (c) shows that for manual workers, the cross-sectional wage profiles are still growing throughout the early 40s for all decades beginning in the 1980s. Therefore, the flat spot of the human capital cannot occur earlier than the early 40s. I set the flat spot age range to ages 46 to 55. This is also consistent with the educational composition of manual workers, with average schooling over the sample period slightly above 12 years (see Figure F.1) and who are dominated by workers with a high school diploma. This age range aligns with the flat spot range of 46–55 that Bowlus

and Robinson (2012) set for high school graduates.

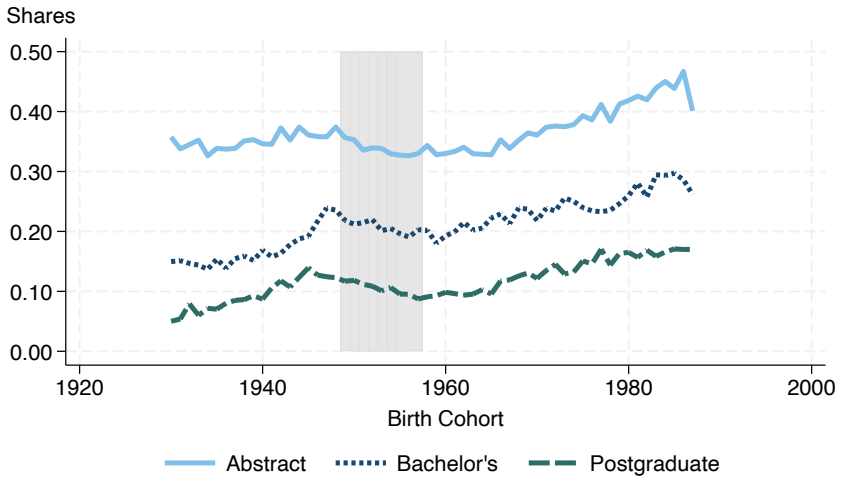
A potential concern is that the cross-sectional wage profile for manual workers in the 1970s peaks much earlier—around age 32—and then declines through the 40s and 50s, in contrast to the flatter or rising profiles observed after 1990. As Fang and Qiu (2023) note, this is the expected signature of a large positive cohort effect. When later cohorts enter the labor market with substantially higher human capital than earlier ones, the cross-sectional profile in any given year can peak at younger ages, even when the underlying experience-human capital profile continues to rise into mid-career. The 1970s and 1980s were the period of the fastest schooling expansion among manual workers in my sample (Figure 2). The share of high school dropouts in manual occupations fell from nearly 50% in 1971 to 21.5% in 1990, with the rate of decline slowing substantially afterwards. Over the same period, the share of workers with some college education rose from 13.2% to 26.6%, also stabilizing afterwards. Large inter-cohort gains in human capital during these decades can thus account for the early-peaking cross-sectional wage profiles of the 1970s and their gradual disappearance in later cross-sections.

Additional evidence supporting this assumption comes from longitudinal data. Böhm et al. (2024) estimate a structural model of occupation-specific skill accumulation using German administrative panel data, allowing skill accumulation profiles to differ across occupations. Their estimates show that skill accumulation is concave in all occupational groups but that the age at which it slows and eventually ceases differs systematically across broad occupational categories, consistent with my choice of flat-spot age ranges. In managerial, professional, and technician occupations (my abstract group), skill accumulation slows with age but remains positive through age 54, supporting my use of a later flat spot for abstract occupations. By contrast, in service and care occupations (which broadly correspond to my manual group), skill accumulation is much flatter and is essentially complete by the mid-40s, consistent with my earlier flat spot for manual occupations. Because these results are identified from within-worker changes in longitudinal data, they provide direct evidence that skill accumulation among mature workers is approximately flat over the age ranges used in my analysis.

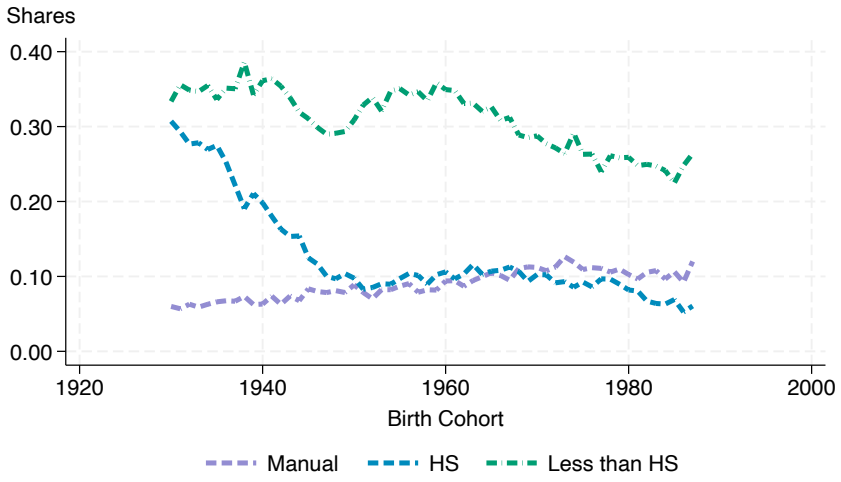
F.3 Flat Spot in Routine Occupations

The challenge with routine occupations is that the direction of cohort quality changes is theoretically ambiguous. The trend of rapid employment growth in both high-skill abstract and low-skill manual jobs has substantially reduced the share of employment accounted for by middle-skill routine jobs (Acemoglu and Autor 2011). However, unlike manual and abstract occupations, routine occupations face both outflows to higher-skill and outflows to lower-skill occupations. Cortes (2016) shows that workers exiting routine occupations sort heterogeneously: routine cognitive workers tend to transition upward to abstract occupations, while routine manual workers are more likely to transition downward to non-routine manual occupations. This bidirectional sorting makes it difficult to sign the net cohort effect as the occupation may experience both positive selection (lower-ability workers leave for manual jobs) and negative selection (higher-ability workers leave for abstract jobs) simultaneously.

Given this ambiguity, I cannot rely on cohort composition arguments to identify the flat spot age range for routine workers. Instead, I use two complementary approaches. First, routine workers have similar educational attainment to manual workers (12 years on average, see Figure F.1), suggesting similar age of entry to the labor market. Second, the nature of routine skills suggests they should not depreciate earlier than purely manual skills (which require motor skills that deteriorate near retirement) nor accumulate as slowly as abstract skills (which require extensive training and experience). These considerations point to a flat spot age range similar to manual workers. Therefore, I set the baseline flat spot age range for routine workers at 46–55, and explore the robustness of estimated price series to alternative specifications.



(a) Abstract occupations and higher education



(b) Manual occupations and high school or less

Figure F.2: Share of Workers Aged 35–40 by Occupation and Education

Notes: This figure shows the share of workers aged 35–40 in the MCPS by birth cohort. Panel (a) reports the share of workers employed in abstract occupations and the share holding a bachelor's or postgraduate degree. Panel (b) reports the share of workers employed in manual occupations and the share with a high school degree or less. The sample includes full-time, full-year male wage and salary workers in the MCPS 1971–2023.

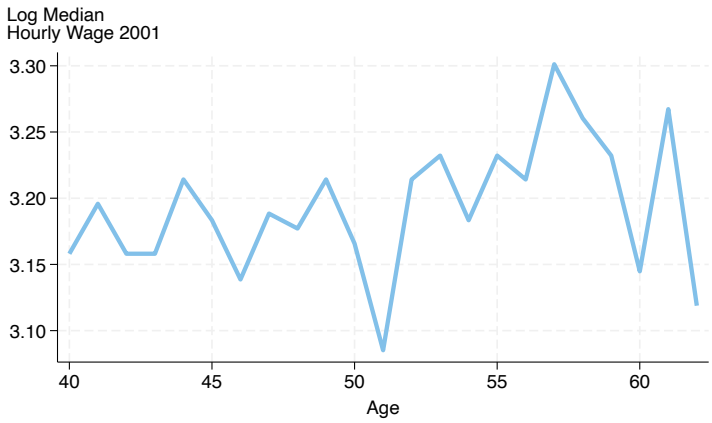
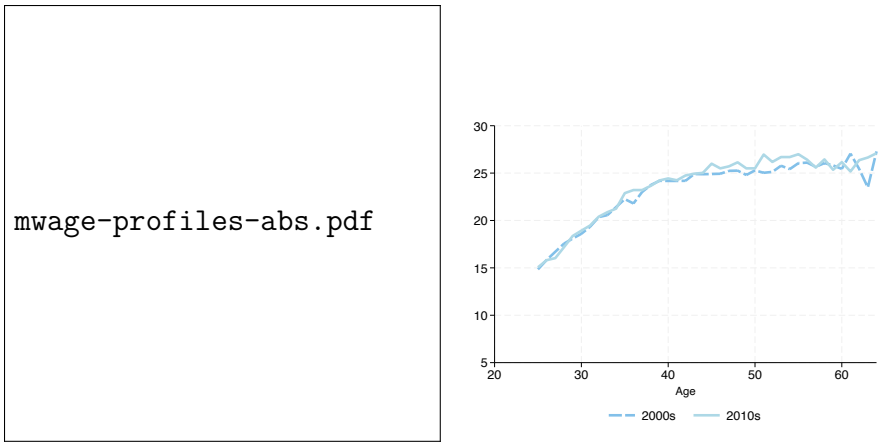
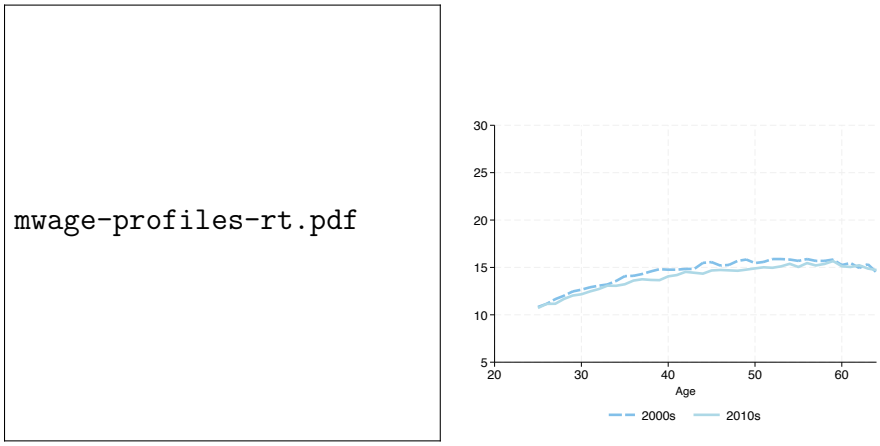


Figure F.3: Log Median Hourly Wage Profile in 2001, by Age

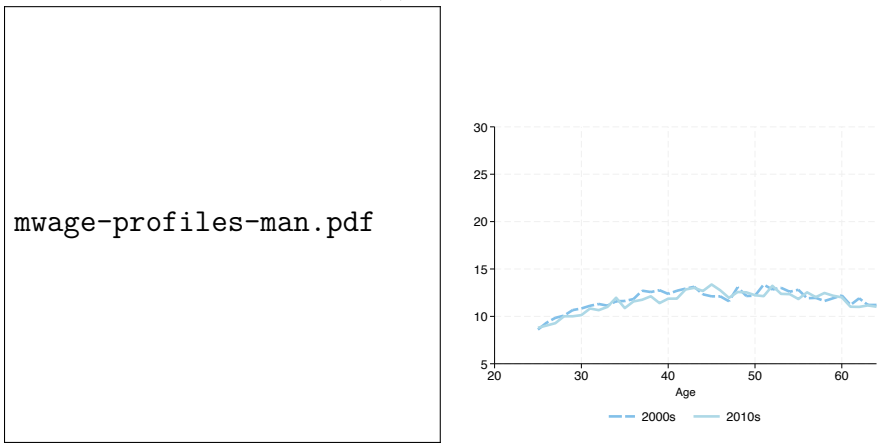
Notes: This figure shows the log median hourly wage for workers in the MCPS 2001 who are employed in abstract occupations, by age. The sample includes full-time, full-year male wage and salary workers in the MCPS 1971–2023. Median hourly wages are computed using CPS sampling weights.



(a) Abstract



(b) Routine



(c) Manual

Figure F.4: Cross-sectional wage profiles by occupational group

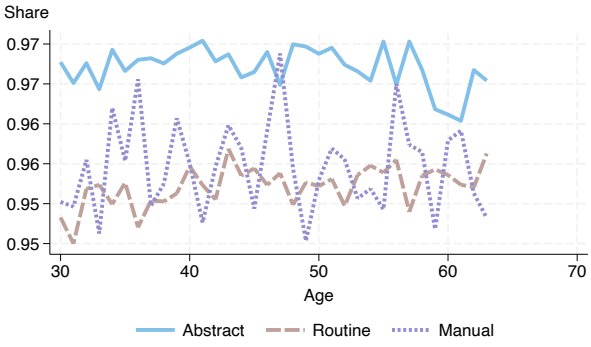
Notes: This figure shows cross-sectional wage profiles by occupational group within 10-year bins. The sample includes full-time, full-year male wage and salary workers in the MCPS, 1971–2023. Median hourly wages are computed using CPS sampling weights.

G Differences in Contracts Across Occupations

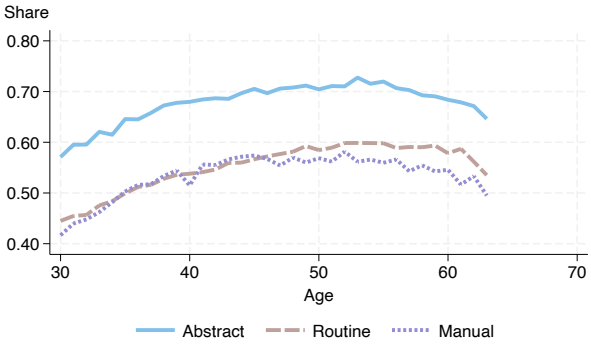
The March CPS asks respondents whether their employer or union paid for all or part of the premiums for an employment-based group health insurance plan in the previous year. Panel (a) of Figure G.1 shows the average share of workers with such coverage across the life cycle. Coverage rates are uniformly high across all occupational groups, approaching 95%, suggesting that differences in total compensation attributable to health insurance are likely to be small.

Panel (b) of Figure G.1 shows the average share of workers enrolled in a workplace pension plan across the life cycle. Abstract workers are substantially more likely to be enrolled in a pension plan than routine or manual workers, and manual workers approaching retirement are slightly less likely to be enrolled than their routine counterparts. This disparity in non-wage compensation is relevant for the interpretation of the estimated price series. If employers partially compensate workers for the absence of a pension plan through higher wages at older ages, the flat spot wages used to identify prices would reflect total compensation adjustments rather than pure prices, introducing a bias whose direction and magnitude depend on how the wage offset varies across occupational groups and over time.

Table G.1 reports cumulative price changes for each occupational group, comparing estimates from the full sample with those restricted to workers covered by an employer-provided health insurance plan or enrolled in a workplace pension plan. For routine and manual occupations, the restricted and full-sample estimates are closely aligned, suggesting that benefit coverage does not systematically affect price identification in these groups. For abstract occupations, however, the restricted sample yields price growth instead of a small decline. This result suggests that the baseline estimates of price growth in abstract occupations are conservative: accounting for non-wage compensation differences reinforces the conclusion that abstract prices have risen relative to routine and manual prices.



(a) Group Health Plan



(b) Pension Plan

Figure G.1: Share of Workers with Employer-sponsored Pension and Group Health Plans by Age

Notes: This figure shows the average share of workers who report having access to employer-sponsored group health plans or pension plans by age. The sample includes full-time, full-year male wage and salary workers in the MCPS 1980–2023.

Table G.1: Cumulative Price Change by Occupation and Sample, 1980–2022

	Full Sample	With Health Plan	With Pension Plan
Abstract	-2.7%	8.8%	9.0%
Routine	-21.0%	-20.9%	-19.9%
Manual	-22.4%	-28.4%	-24.7%

Notes: Entries are cumulative percentage changes in the price index between 1980 and 2022. Prices are estimated using the flat spot method with baseline flat spot windows: ages 51–60 for Abstract; ages 46–55 for Routine and Manual. The restricted samples include workers with health plan coverage or pension entitlement respectively. The sample includes full-time, full-year male wage and salary workers in the MCPS 1980–2023.

H Occupational Group Switching Behavior

If senior workers actively switch occupations, the assumption of stable and homogeneous occupational human capital is unlikely to hold. I use the longitudinal component of the Merged Outgoing Rotation Group (MORG) data from 1986 to 2022 to analyze the role of occupational group switching behavior in price estimation. The MORG samples consist of households who answer additional labor market questions in months four and eight in the Basic Current Population Survey (CPS). The responses in month four and month eight can be matched, providing information about changes in labor market outcomes over the year of participation in the survey. I restrict the MORG sample to full-time male wage and salaried workers ages 40 to 64 years, who have reported positive earnings and hours worked and had a valid record of occupation.[31]

I find that broad occupational group choice is highly persistent across both age and time. Figure H.1 reports occupational transitions over a one-year period for full-time workers employed in abstract, routine, and manual occupations in month four of the sample. Around the flat-spot age range, 85–87% of workers in abstract occupations remain in the same broad occupational group one year later. Persistence is similarly high for routine occupations (84–89%), while manual occupations exhibit somewhat greater mobility, with 75–87% of workers remaining in the same occupational group.

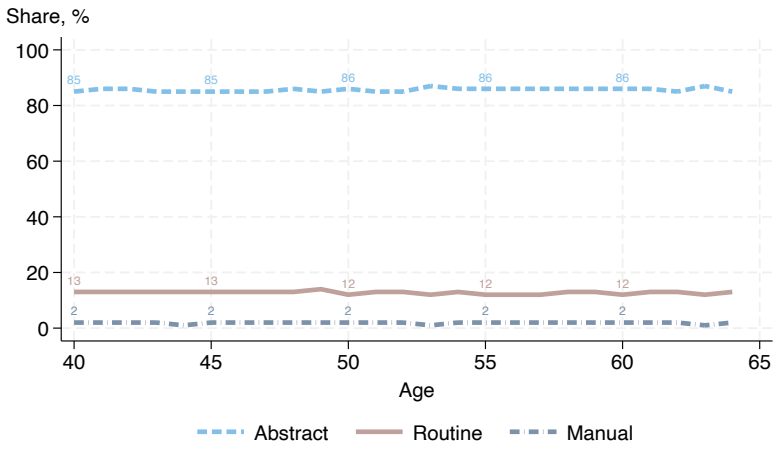
Figure H.2 shows that these patterns are stable over time. Across years, the share of workers remaining in the same broad occupational group ranges from 82–88% for abstract occupations, 86–91% for routine occupations, and 73–84% for manual occupations. Similarly to the life-cycle occupational switching rates, manual occupations exhibit the highest rate of occupational switching, reflecting the smaller size of this occupational group. Nevertheless, persistence remains sufficiently high that occupational switching is unlikely to materially affect the estimates. Appendix I examines this issue directly by restricting the estimation sample to workers who remain in the same broad occupational group.

Figure H.3 analyzes the occupational composition of workers by age and birth cohort. It shows that workers tend to begin their careers in routine or manual occupations and switch to abstract occupations at later stages of their careers. Although the series tend to fluctuate

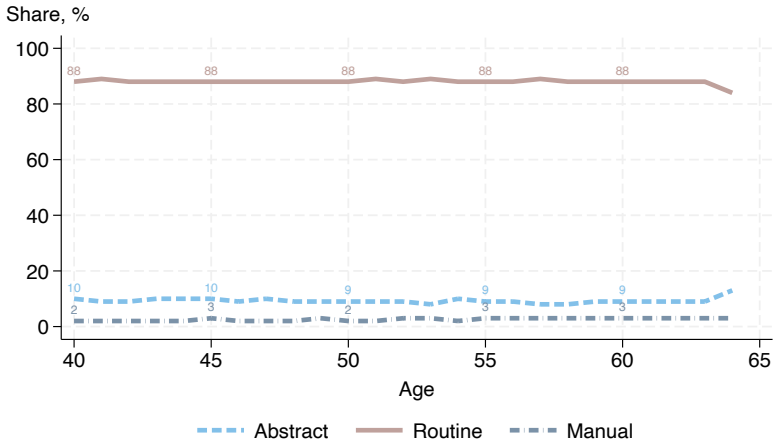
[31] Full-time employment is defined as working 35-plus hours per week.

due to limited sample sizes, the occupational structure for all cohorts seems to become more stable by the age of 40. [32]

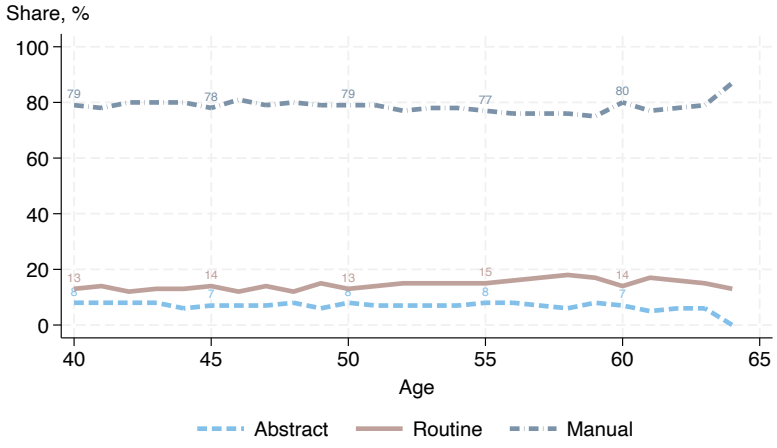
[32] The increase in the share of employment in abstract occupations for workers in their 20s is also a result of abstract workers taking longer to complete their education.



(a) Abstract



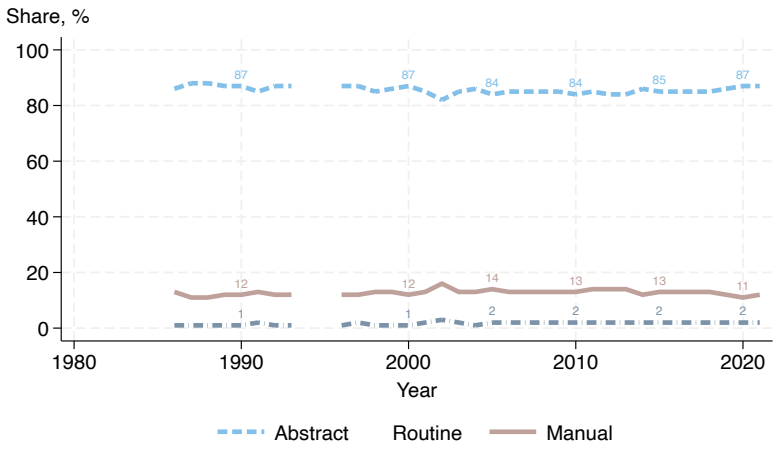
(b) Routine



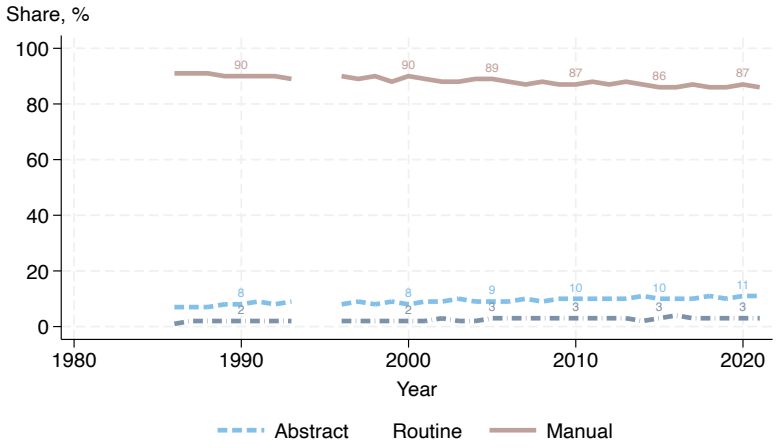
(c) Manual

Figure H.1: Occupational Switching Patterns by Age

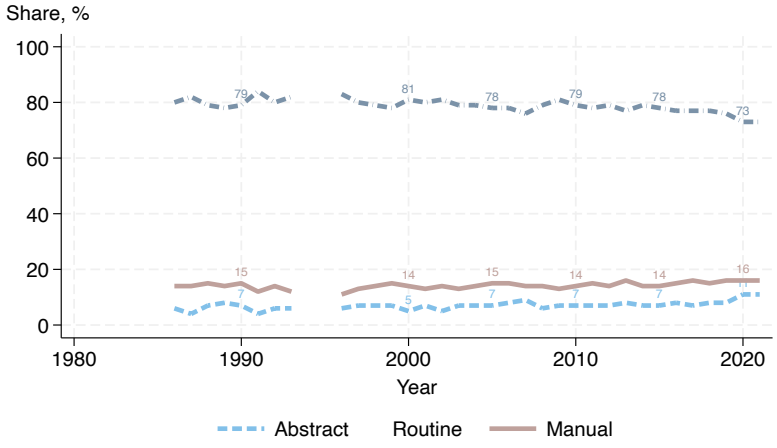
Notes: This figure shows the annual pattern of occupational switching between months 4 and 8 in MORG by age. The sample includes full-time wage and salaried male workers aged 40–64 years from MORG (1986–2022). The sample excludes observations from 1994 and 1995.



(a) Abstract



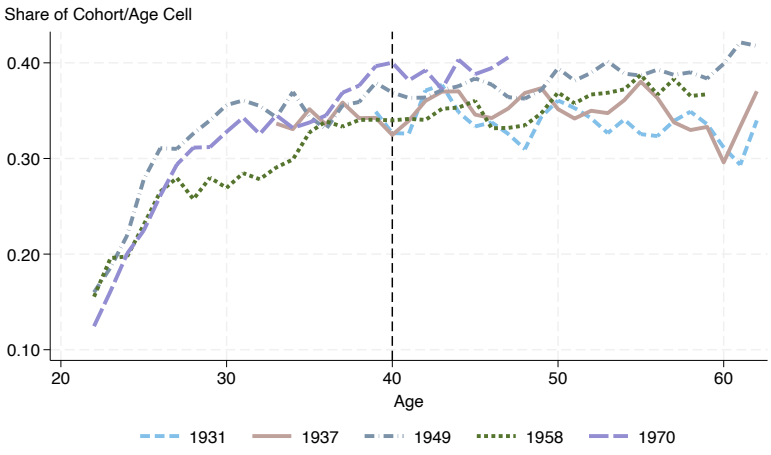
(b) Routine



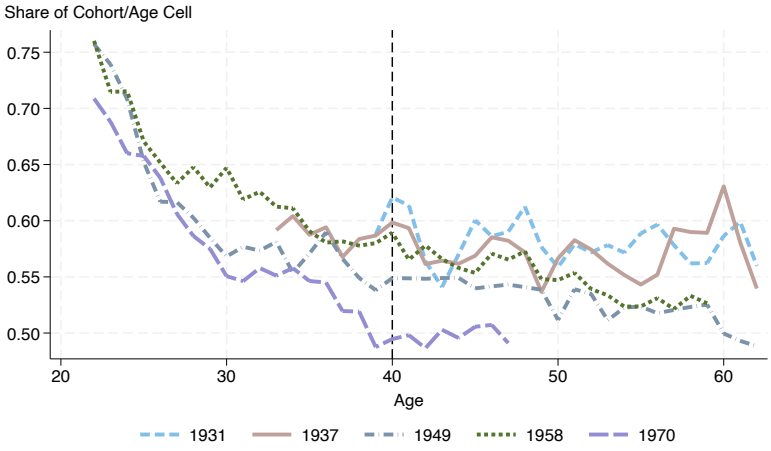
(c) Manual

Figure H.2: Occupational Switching Patterns Over Time

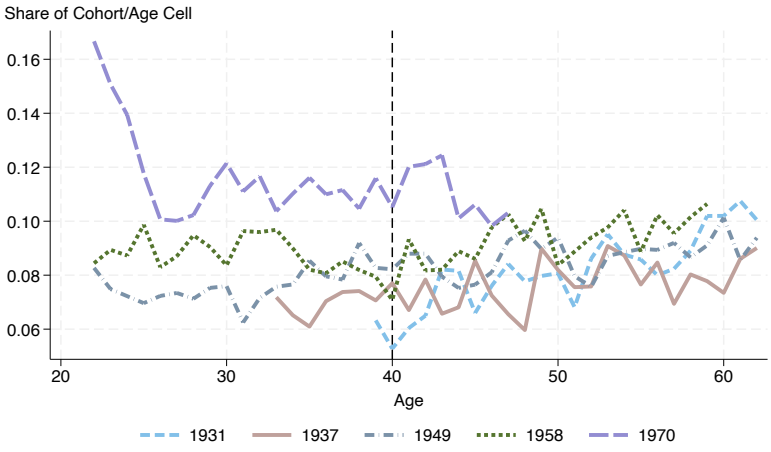
Notes: This figure shows the annual pattern of occupational switching between months 4 and 8 in MORG over time. The sample includes full-time wage and salaried male workers aged 40–64 years from MORG (1986–2022), excluding 1994 and 1995.



(a) Abstract



(b) Routine



(c) Manual

Figure H.3: Occupational Sorting by Birth Cohort

Notes: This figure shows the occupational composition of workers by age and birth cohort. The sample includes full-time, full-year male wage and salary workers in the MCPS 1971–2023.

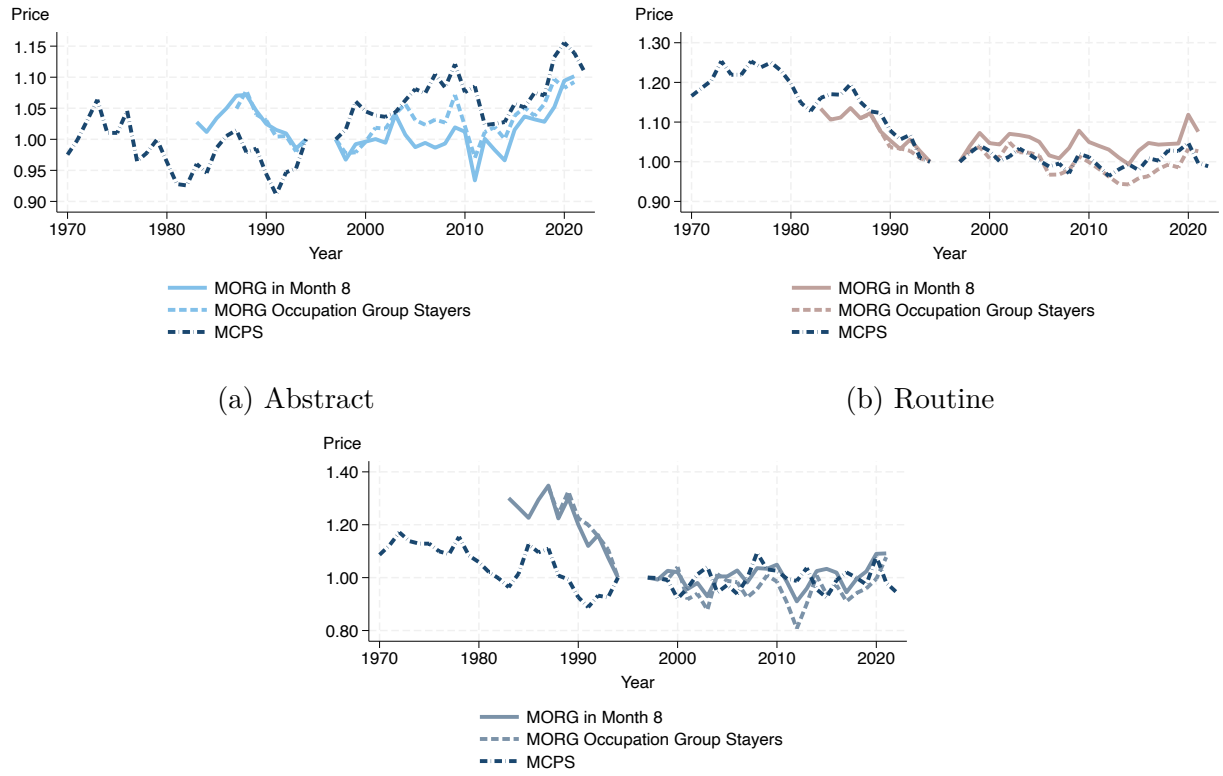
I Sensitivity of Price Estimation to Occupational Switching

To assess whether occupational switching behavior significantly alters my results, I estimate the price series for workers who remained in the same occupational group throughout their participation in the MORG survey. Figure I.1 illustrates the price series estimated for both restricted and unrestricted samples. The unrestricted samples include full-time wage or salaried workers in MORG during month eight of the CPS. The restricted sample includes full-time wage and salaried workers in MORG who stayed in the same occupational group over the year. I additionally include the baseline price series using the full-time full-year sample of male workers in the MCPS.

In 1985 and 1995 the CPS changed its housing unit numbering schemes, preventing matching across some individuals between 1984–1985, 1985–1986, 1994–1995, and 1995–1996. Therefore, I estimate the evolution of occupational human capital prices from 1987 to 1994 and from 1997 to 2022, when occupation stayers can be identified in the MORG sample.

The trends in human capital prices are similar for both occupational group stayers and unrestricted samples across all occupations. This implies that occupational switching has a limited effect on the price estimation results. Price series exhibit a price decline from 1987 to 1994 for all occupations and all samples. Since the late 1990s, prices for routine and manual occupations have been decreasing at a smaller rate compared to the late 1980s and early 1990s. Moreover, prices for abstract occupations have been slowly increasing in both samples.

The MCPS and MORG price series track each other closely from 1997 onward across all three occupational groups, despite being constructed from different samples and different wage measures: the MCPS series divides annual earnings by annual hours, while the MORG series uses the point-in-time weekly earnings reported in month eight. That two independent measures yield the same price trends over this period indicates that the estimated trends are not an artifact of how I construct wages in the MCPS. The two sources differ in magnitude over the earlier 1987 to 1994 window, where the MORG series for abstract and manual occupations declines more steeply than the MCPS series. The direction of the price change is the same in both sources throughout the overlap: routine and manual prices decline and



(a) Abstract

(b) Routine

(c) Manual

Figure I.1: Sensitivity of Price Series to Occupational Switching

Notes: This figure shows the estimated baseline price series alongside price series for full-time workers in MORG (based on wage data from month 8) and full-time occupation stayers in MORG (based on wage data from month 8). Price series use median hourly wages computed with MORG sample weights.

the abstract price is flat to rising. For the routine and manual groups, the comparison gives no indication that the MCPS baseline overstates the price decline that my results turn on. If anything, the decline in the price of manual human capital might have been stronger than MCPS estimates suggest.

J Minimum Wage

One potential concern with the price estimation methodology is that price and quantity estimates may be contaminated by changes in minimum wage legislation. A key advantage of using median wages in the estimation is that it substantially limits this concern. Figure J.1 plots the maximum effective minimum wage against median hourly wages for the abstract, routine, and manual occupational groups over the sample period. The effective minimum wage for each observation is defined as the higher of the applicable state minimum wage and the federal minimum wage, deflated to 2000 U.S. dollars using the CPI; state and federal minimum wage series are drawn from the Federal Reserve Bank of St. Louis (Federal Reserve Bank of St. Louis 2026). The maximum effective minimum wage series is then the yearly maximum of these values across all observations in the sample. As the figure shows, median wages used in both the price and quantity estimation lie well above the maximum effective minimum wage in all three occupational groups throughout the sample period. Furthermore, the flat spot identification relies exclusively on wages of workers aged 46–55 (51–60 for abstract occupations), a group that is substantially less likely to be employed in minimum-wage jobs than the broader working-age population.

Nevertheless, minimum wage floors may still compress the lower tail of the wage distribution, particularly in the manual occupational group, which contains the lowest-paid workers and a higher share of service occupations subject to minimum wage legislation. As a robustness check, I re-estimate the price series for the manual and routine occupational groups following the methodology described in Appendix E, adding a continuous control for the effective state minimum wage to the quantile regression used to estimate median hourly wages for each occupation–age–year cell.

State identifiers are not reliably available in the 1970s because multiple states are grouped together, and these groupings change over time; therefore, I focus on the price change since

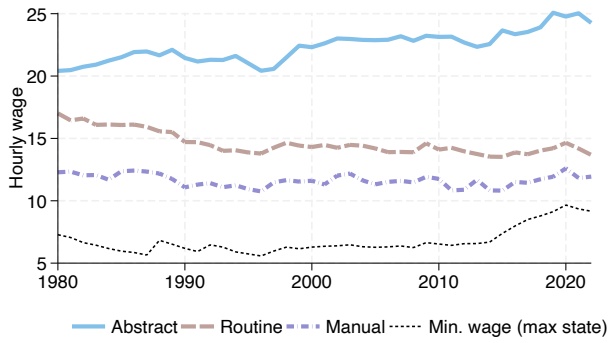


Figure J.1: Maximum effective minimum wage and median wages by occupation group

Notes: This figure plots the maximum effective minimum wage against median hourly wages for the abstract, routine, and manual occupational groups. The effective minimum wage for each observation is the higher of the applicable state and federal minimum wage; the maximum effective minimum wage series is the yearly maximum of this value across all observations in the sample. All wages, including the minimum wage series, are deflated to 2000 U.S. dollars using the CPI. State and federal minimum wage series are from the Federal Reserve Bank of St. Louis (2026). The sample includes full-time, full-year male wage and salary workers aged 25–64 years in the MCPS, covering earnings years 1980–2022. Median hourly wages are computed using CPS sampling weights.

1980 (Flood et al. 2024). Adding the effective state minimum wage as a control to the quantile regressions used to estimate the price series increases the estimated decline in manual occupational prices between 1980 and 2022 by roughly 13 log points, while reducing the estimated decline for routine occupations by about 2 log points. The larger estimated decline for manual occupations is consistent with minimum wage increases having raised wages in manual occupations, thereby attenuating the measured decline in the baseline estimates. These results suggest that the baseline estimates are conservative and that the long-run deterioration in manual occupational prices may have been even larger than the headline estimate implies.

K Prices for College Graduates in Abstract and Routine Occupations

Although Bowlus and Robinson (2012) found that human capital prices followed similar trends across education, my findings suggest that human capital price trends diverged for workers employed in abstract compared to routine and manual occupations. While education is an important characteristic of workers' human capital, the growing wage inequality among college-educated workers underscores the role of other characteristics, including workers'

occupations, in explaining the change in wage inequality (Lemieux 2006).

To explore whether defining workers' human capital by occupation enhances the understanding of changes in wage inequality across college workers, I estimate changes in prices for an efficiency unit of human capital for workers with college and above employed in abstract and routine occupations. Following Bowlus and Robinson (2012), I use a subsample of full-time, full-year male workers ages 50 to 59 with bachelor's or postgraduate degrees to estimate the price series, but I estimate separate price trends for workers employed in abstract and routine occupations.[33]

Figure K.1 shows that the price for workers in abstract occupations has remained relatively stable since the mid-1990s. By contrast, the price for workers in routine occupations has been steadily declining. Although the estimated price series has limited statistical power due to the reduced sample size after partitioning, the results are consistent with the idea that wages of workers with the same education level may be differentially affected by changes in the labor market depending on the tasks they perform in the workplace.

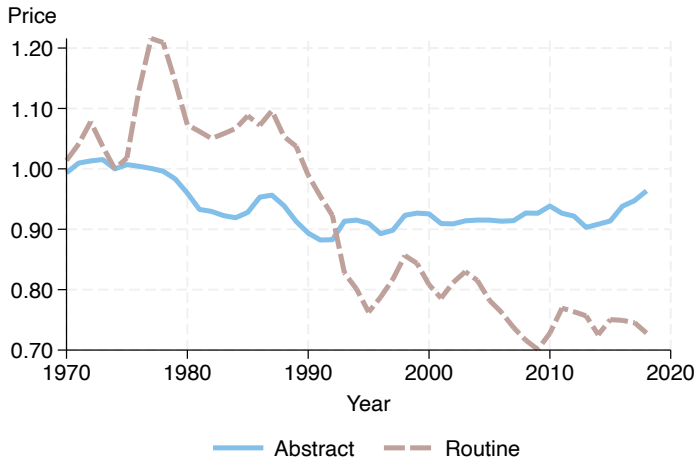


Figure K.1: Price Series for College-Educated Workers in Abstract and Routine Occupations

Notes: This figure shows the evolution of estimated prices of human capital for workers with bachelor's or postgraduate degrees employed in abstract and routine occupations. The sample includes full-time, full-year male wage and salary workers aged 50–59 years in the MCPS. The plotted series are smoothed using a centered three-year moving average.

[33]The analysis for college-educated workers in manual occupations was excluded due to insufficient sample sizes.

L Prices of Social Tasks

A growing literature highlights the increasing importance of social skills in the labor market and documents a substantial increase in returns to social skills over recent decades (Deming 2017b; Edin et al. 2022). Moreover, social skills have been shown to be particularly important for less-educated workers (Cnossen et al. 2025). The wage gains in occupations intensive in social tasks are naturally interpretable as reflecting increases in the price of these tasks, increases in the quantity of social human capital, or both. In this appendix, I apply the flat-spot method to occupational groups defined by social skill intensity to assess changes in the price of social-task-intensive occupations.

To divide workers into subsamples employed in occupations with high and low intensity of social tasks, I use the social task intensity index constructed by Deming (2017b) from O*NET data (Deming 2017a). The index is merged to my main sample by occupation code, and I classify workers as employed in high- or low-social-task occupations according to whether their occupation's index value falls above or below the median. This classification cuts across my baseline taxonomy. Although social-task-intensive occupations are concentrated in the abstract group, they also appear in routine and manual occupations (see Table L.1).

Table L.1: Distribution of Occupational Groups by Social Skill Intensity

	Low Social Skills	High Social Skills
Abstract	6.5	63.9
Routine	83.2	27.7
Manual	10.3	8.4
N	663971	649483

Notes: This table reports the composition of workers employed in occupations with high and low social task intensity. Workers are classified as employed in high- or low-social-task occupations according to whether their occupation's social task intensity index lies above or below the median. The social task intensity measure is taken from the replication package for Deming (2017b). The sample includes full-time, full-year male wage and salary workers in the MCPS.

Figure L.1 reports price series estimates for workers employed in occupations with high and low intensity of social tasks. Since high-social-skill occupations are concentrated among abstract workers and low-social-skill occupations among routine and manual workers (Table L.1), I use the benchmark flat spots corresponding to these baseline groups: ages 51–60

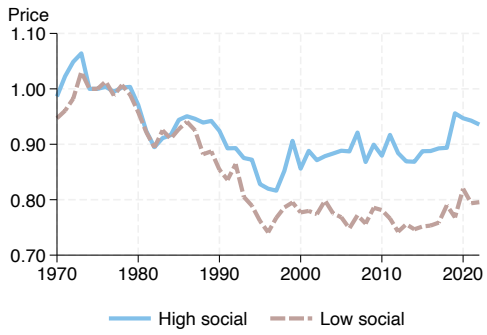


Figure L.1: Price Series by Social Task Intensity

Notes: This figure shows the evolution of estimated prices of human capital for workers employed in occupations with high and low social task intensity. Workers are classified as employed in high- or low-social-task occupations according to whether their occupation's social task intensity index value falls above or below the median. The social task intensity measure is taken from the replication package for Deming (2017b). The price series are estimated using the flat-spot method with benchmark flat spot ages 51–60 for high-social-task workers and 46–55 for low-social-task workers, corresponding to the flat spot ranges used for abstract and routine/manual workers in the baseline estimation. The sample includes full-time, full-year male wage and salary workers in the MCPS.

for high-social-skill workers and ages 46–55 for low-social-skill workers. This allows a direct comparison of price trends estimated under the social skill taxonomy with those obtained under the baseline abstract, routine, and manual classification.

When occupations are divided by social task intensity, prices decline for both high- and low-social-task groups, with the gap between them emerging in the late 1990s. This contrasts with the baseline results, where the price for abstract occupations increases in absolute terms while prices for routine and manual occupations decline. The social task taxonomy obscures this heterogeneity because high-social-task occupations include workers from routine and manual groups alongside abstract workers (see Table L.1). The resulting price series reflects a mixture of price trajectories that masks the absolute price gains of abstract workers identified under the baseline classification.

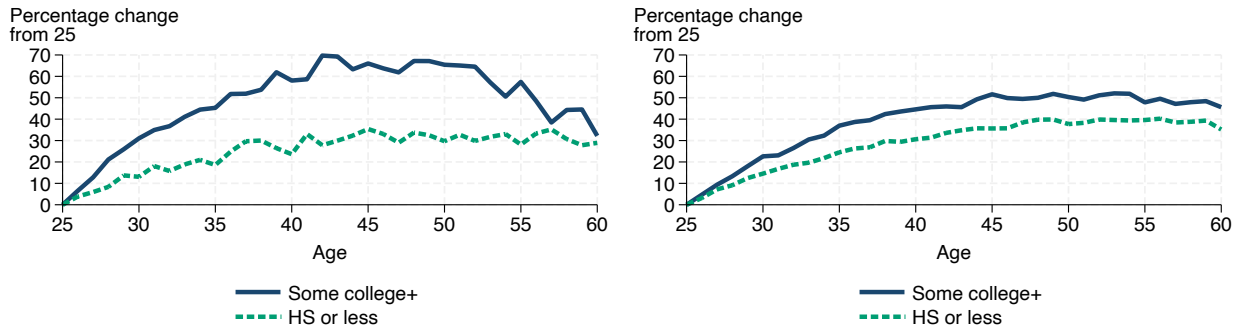
M Human Capital Growth in Manual and Routine Occupations

Figure M.1 shows the life-cycle profile of human capital in manual occupations, separately for workers with at least some college education and workers with a high school degree or less. Workers with at least some college accumulate human capital substantially faster early

in their careers. Their median efficiency units rise by almost 70% between age 25 and the early 40s, compared to about 30% for workers with a high school degree or less. This within-occupation education gradient in human capital accumulation provides the channel through which the compositional shifts documented in Figure 2 translate into the rotation of cohort profiles shown in Panel (c) of Figure 6. As later cohorts of manual workers entered the labor market with higher educational attainment, a growing share of them followed the steeper accumulation path of workers with at least some college, changing the average life-cycle profile of human capital within manual occupations.

By contrast, the two education groups follow much more similar life-cycle profiles in routine occupations. Workers with at least some college still accumulate slightly more human capital over the life cycle than workers with a high school degree or less, but the gap between the two profiles is far smaller than in manual occupations. One interpretation is that the codified, rule-based nature of routine tasks limits the on-the-job returns to general cognitive skill. Because routine tasks are well-defined and procedural by construction (Acemoglu and Autor 2011), additional years of schooling provide a smaller advantage in subsequent human capital accumulation than in manual occupations, where task content varies more substantially across jobs and includes non-codifiable elements such as interpersonal judgment and problem-solving. The relatively weak education gradient in routine accumulation helps explain why routine cohort profiles do not exhibit the rotation observed in manual occupations (Panel (b) of Figure 6).

Figure M.2 brings the upskilling story down to the level of individual jobs. It plots the log-point increase in the quantity of human capital between 1970–1974 and 2018–2022 against the percentage-point increase in the share of workers with at least some college, with one marker per detailed occupation within the manual group. Because partitioning into detailed occupations leaves small cells and noisy estimates, I restrict attention to occupations with at least five observations in each year and at least three qualifying years out of the five in each window, and I average across the qualifying years for each occupation. The occupations line up along an upward slope: where the share of workers with at least some college rose most, the quantity of human capital rose most as well, and the pattern holds across protective, food and cleaning, and personal services alike. The share of workers with at least some college



(a) Manual (b) Routine

Figure M.1: Life-cycle human capital profiles by education in manual and routine occupations

Notes: This figure shows the percentage change in median efficiency units of human capital from age 25, by age, for workers in manual (Panel (a)) and routine (Panel (b)) occupations, separately for workers with at least some college education and workers with a high school degree or less. Efficiency units are computed as the ratio of hourly wages to the estimated occupation-specific price of human capital. Profiles pool all years and birth cohorts in the sample. The college group includes workers with some college, a bachelor's degree, or a postgraduate degree; the less-than-college group includes high school dropouts and high school graduates. The sample includes full-time, full-year male wage and salary workers in the MCPS 1971–2023.

increased in every manual occupation, but the occupations that upskilled most, such as firefighters, inspectors, and childcare workers, also saw the largest growth in human capital quantity. Occupations with less upskilling, such as barbers, cooks, and hairdressers, saw substantially lower or even negative human capital growth. While these patterns should be interpreted with caution because of sample partitioning and small cell sizes, they are consistent with educational upskilling driving the growth in human capital within manual occupations.

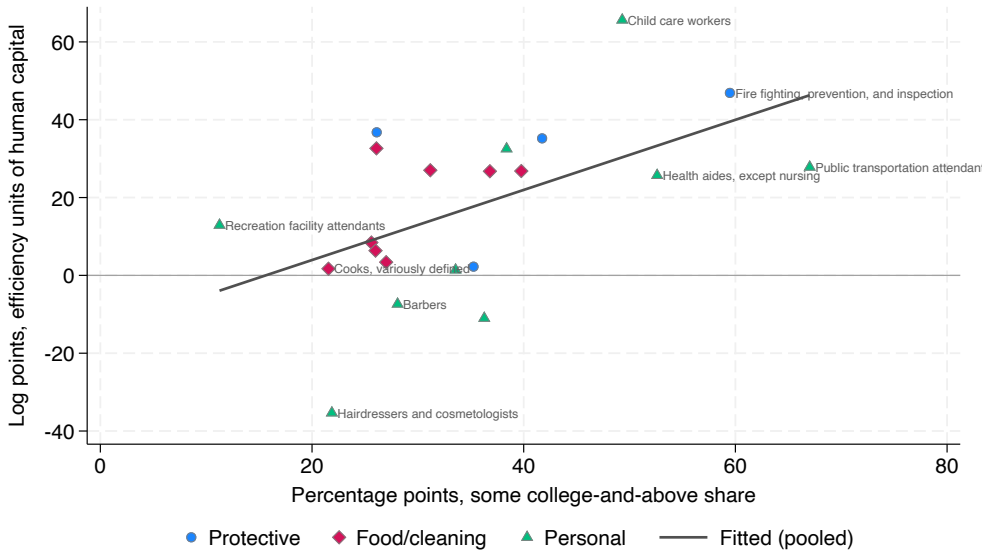


Figure M.2: Change in human capital and educational attainment by detailed manual occupations

Notes: Each marker is a detailed occupation within the manual group. The horizontal axis is the percentage point change in the share of workers with some college or above, and the vertical axis is the log points change in human capital quantity, each computed as the difference between the average over the first five and the last five years of the sample. Only occupations with at least 5 observations in at least 3 of the first five and at least 3 of the last five years of the sample are included. Human capital quantity is the median hourly wage deflated by the price of manual human capital, so the same deflator applies to every manual occupation. Markers are colored by sub-occupation group, and the solid line is the pooled OLS fit across all manual occupations. The three occupations with the largest and the three with the smallest combined change in educational attainment and human capital are labeled. The sample includes full-time, full-year male wage and salary workers in the MCPS 1971–2023.